



FAPS

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Formed coil technology and its application in the automotive powertrain

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

This session provides an overview of the use of formed coil technology in the context of the electrification of the automotive powertrain.

Formed coil technology is an attractive option for the production of high-performance automotive traction machines.

After attending the lecture unit „Formed coil technology and its application in the automotive powertrain" you should:

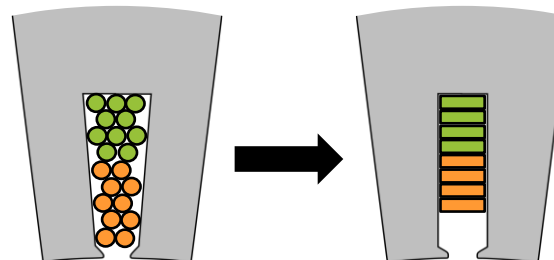
- have understood the key challenges of the electrification of the automotive powertrain,
- be able to name the potential for the use of formed coil technology in the automotive powertrain,
- be able to distinguish between the different types of formed coil and bar windings with their advantages and disadvantages,
- know production chains and technologies for the manufacture of formed coil and bar windings and
- know the advantages and disadvantages of formed coil technology compared to established winding processes.



Graphic: Porsche



Graphic : Risomat



Graphic : ATOP



Graphic: BMW

This session provides an overview of the use of formed coil technology in the context of the electrification of the automotive powertrain.

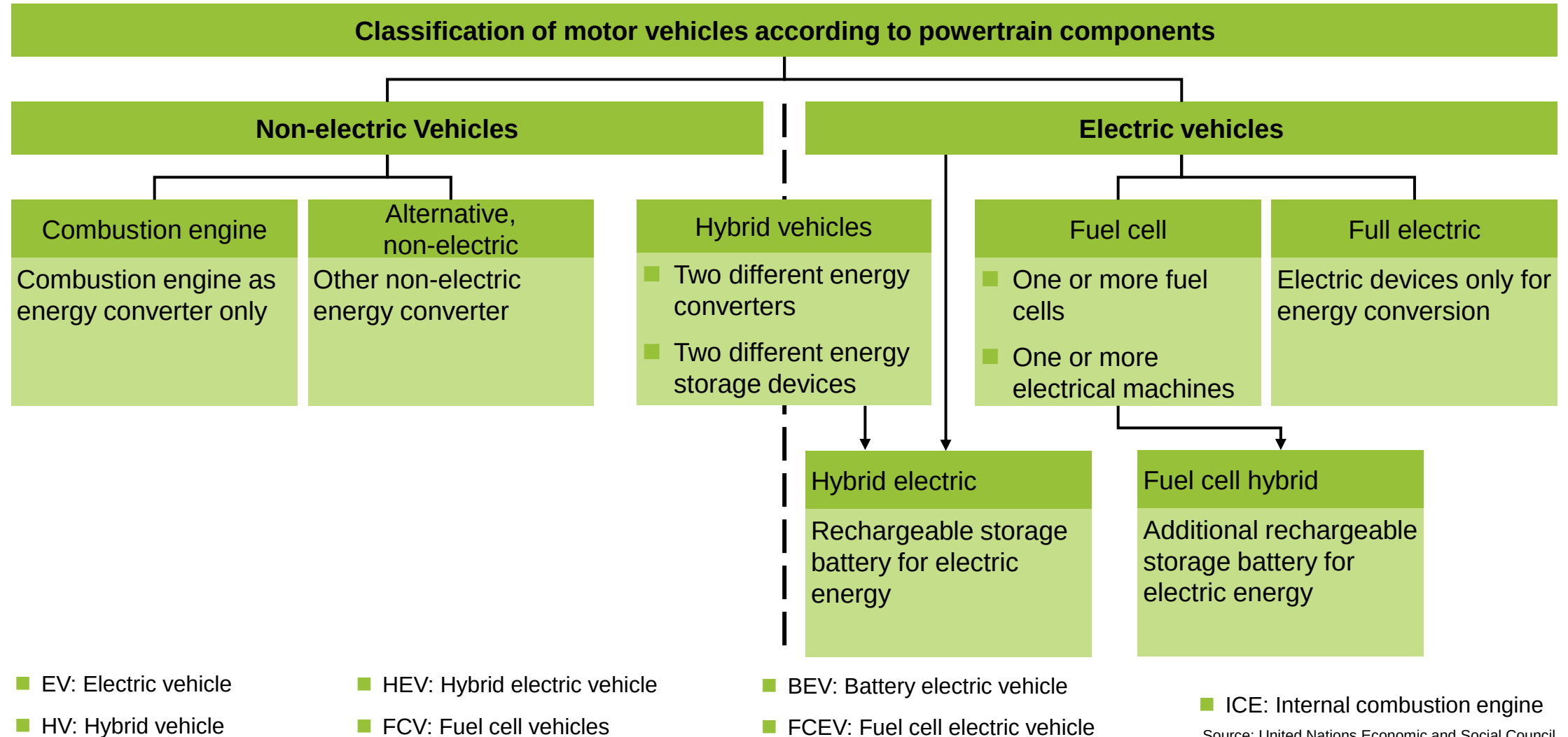
- **Electrification of the automotive powertrain**
- The formed coil and bar technology
- Process chains for the production of formed coil and bar windings
- Process technologies for the production of formed coil and bar windings for automotive traction machines
 - Slot insulation
 - Straightening
 - Stripping
 - Cutting
 - Shaping
 - Assembly
 - Twisting
 - Contacting
 - Insulating



Graphic: ATOP

Electrifying the automotive powertrain occurs in a variety of ways, as defined by the United Nations.

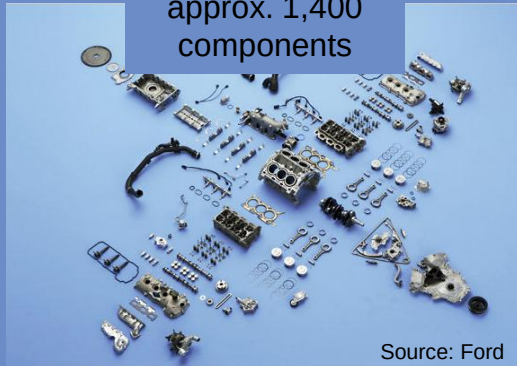
Slide from Lec. 1



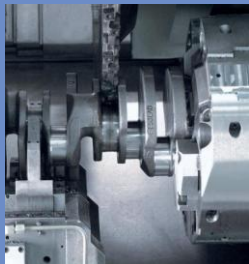
Electric traction drives differ significantly from combustion engines in terms of system complexity and production requirements.

Internal combustion engine

approx. 1,400
components



Source: Ford



Graphics: Audi, EMAG, Böhlinger

Reduction in the number
and complexity of
components

From predominantly machining
production to complex
assembly processes

Electric traction drive

approx. 100 - 200
components



Graphics: Continental, Lehner

Today's industrial drive requirements are very different from the automotive environment.

Comparison of a 100 kW industrial drive with a combustion engine

Criteria	Sector	
	Industry	Automotive
		
Market volume Germany (p.a.)	< 1000 units	> 1 Mio units
Price	10000 €	2000 €
Delivery quality (ppm)	1000	100
Adherence to deadlines	95%	100% (JiT)
Service life	40000 h	4000 - 8000 h
Share of labor costs	25%	5%

The cost and quality targets for electric automotive traction machines can only be achieved if the product and process are developed in parallel.

Product requirements

- Increasing power densities
- Minimization of installation space and weight
- Optimization of the cooling concepts
- Recyclability and reduction or replacement of rare earth metals



Graphic: ZF Friedrichshafen

Process requirements

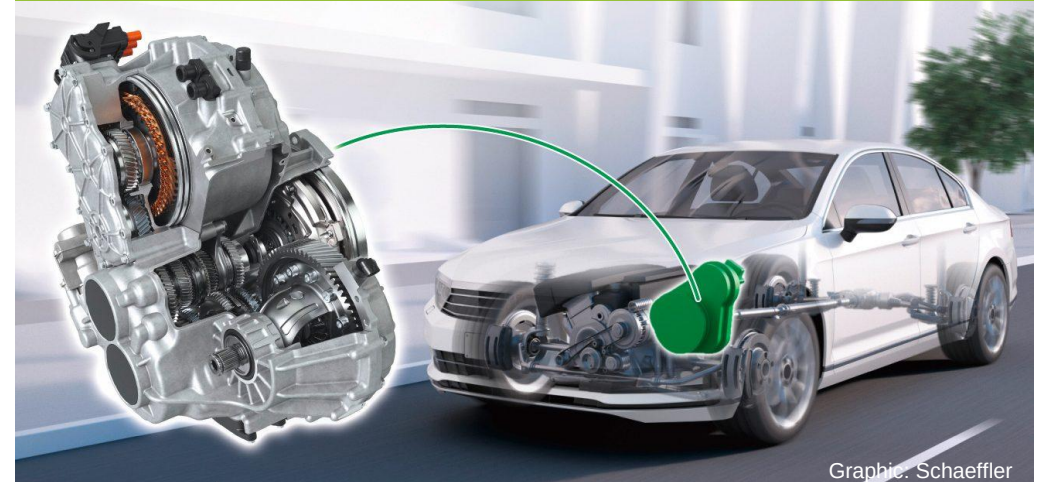
- Leveraging the economies of scale
- Safeguarding the product quality
- Increasing degree of automation while maintaining flexibility
- Optimization and Process integration of testing technologies

Automotive applications place high demands on electric traction machines.

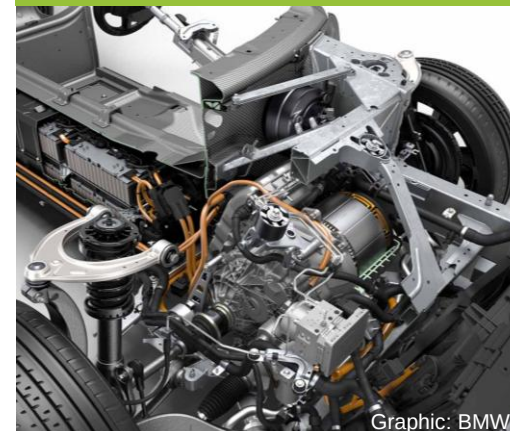
Automotive-specific requirements for electrical machines

- Low manufacturing costs
- Lower service life than industrial drive
- Good controllability (high dynamics)
- High efficiency over a wide torque-speed range
- Optimized cooling concepts
- High thermal overload capacity
- High power density
- Small installation space
- Low mass
- High reliability
- High robustness (vibration, impact)
- Low acoustic emissions
- Easy to integrate into the overall system

Hybrid vehicles



Battery electric vehicles

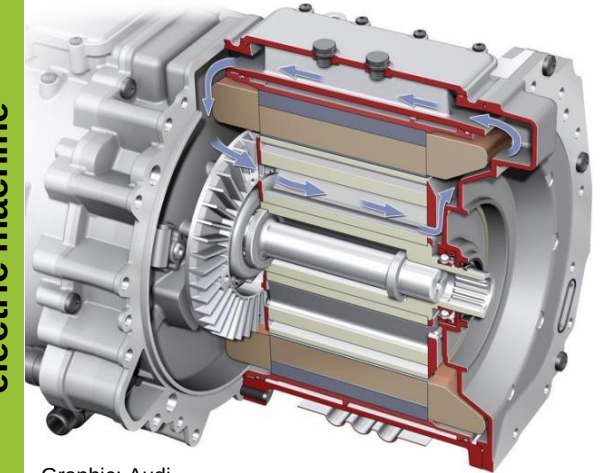


The thermal overload capacity as well as a high gravimetric and volumetric power density characterize an automotive drive.

Thermal overload capacity

- Heating due to winding and iron losses
- Maximum output of the electric motor limited by cooling system and thermal load limit of the enamelled copper wire
- The better the heat dissipation, the longer and the greater the possibility of overloading
- Thermal overload capacity → Higher peak power available for longer

Liquid-cooled electric machine



Graphic: Audi

Installation space / power density

- When integrated into conventional gearboxes, particularly severe axial space limitations → short winding heads (PSM with single-tooth winding)
- Desire for short winding heads and high active length
- Greater degrees of freedom for power-split gearboxes
- High power density due to high copper fill factor and good heat dissipation of the machine (liquid cooling, impregnation)

Disc-shaped PSM with single-tooth winding



Graphic: Bosch

Minimizing the power loss increases the efficiency, which means that longer ranges can be achieved with the same battery capacity.

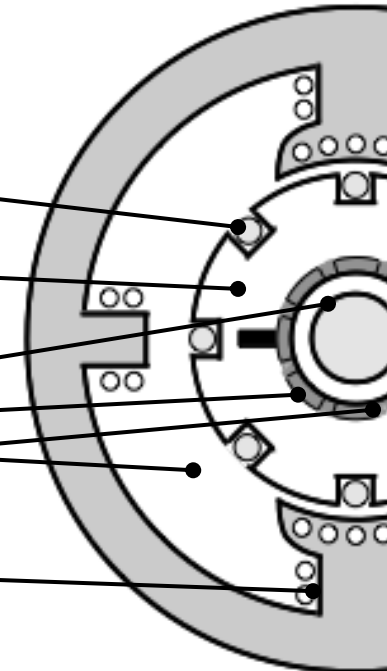
Efficiency η

- The efficiency describes the ratio of power output to absorbed power
- Goal: Minimization of power loss

$$\eta = \frac{P_{output}}{P_{input}} = \frac{P_{input} - P_{loss}}{P_{input}}$$

Significant losses in electrical machines

- Winding losses
 - Joule heating of the winding
 - Current displacement losses
- Losses in the magnetic circuit
- Hysteresis losses
- Eddy current losses
- Friction losses (air and bearing friction, brush friction)
- Brush transition losses (FE SM, DCM)
- Losses of currents used to generate the excitation (FE SM, DCM)



The automotive industry has awakened the electric motor from its slumber.

Design

Drive Unit: Motor | Tech Talks

Video:
Lucid

<https://www.youtube.com/watch?v=U7IHZxNC6hc>



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Graphic: ATOP

Windings for three-phase machines may use one of two types of designs.

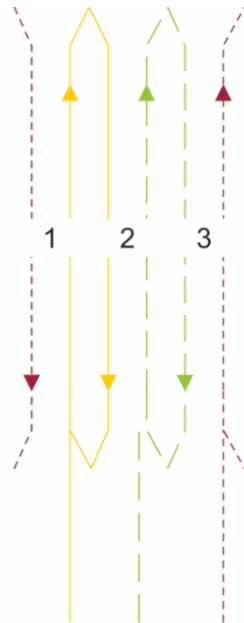
Slide from Lec. 5

Concentric windings

- Coils do not overlap in the winding head
- Single-layer arrangement: one coil side per slot
- Double-layer arrangement: two coil sides per slot

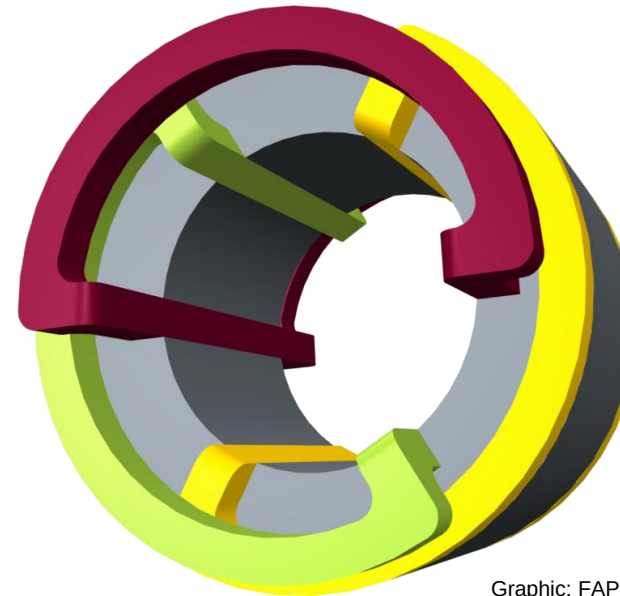


Graphic: FAPS

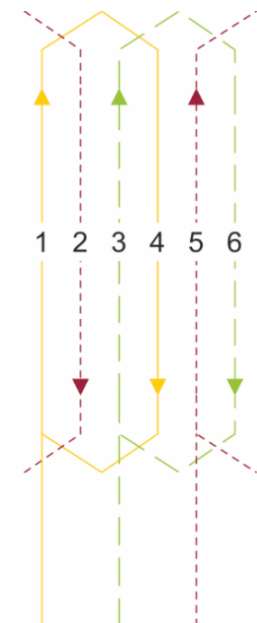


Distributed windings

- Coils overlap each other, distributed over the stator circumference
- Coils “skip” several slots
- Single- and double-layer arrangements possible

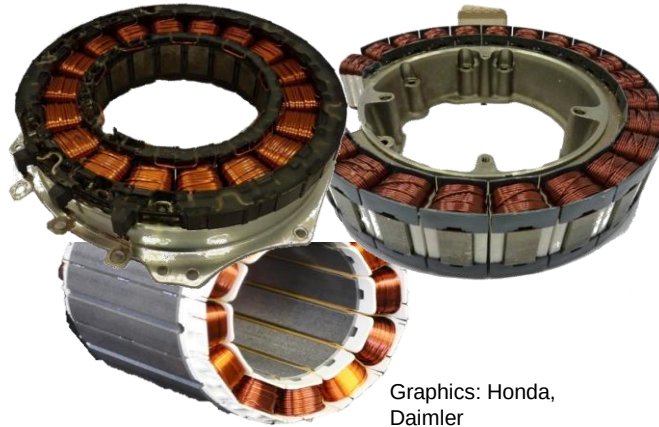


Graphic: FAPS



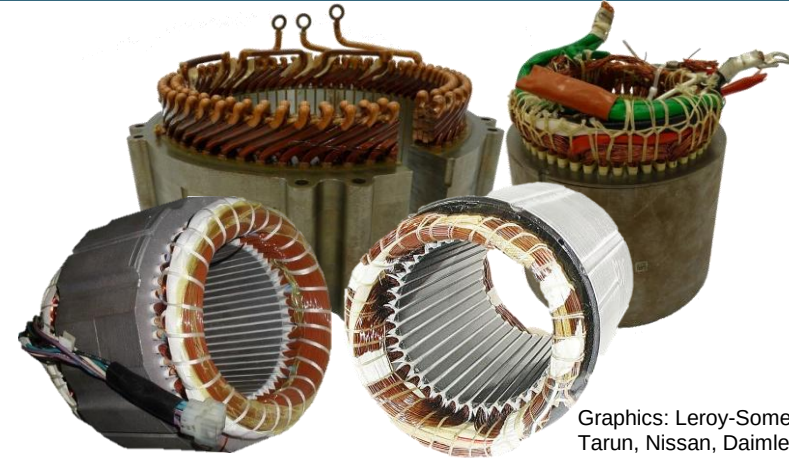
Distributed windings are more advantageous during motor operation, however concentric windings are significantly easier to manufacture.

Concentric windings



Graphics: Honda, Daimler

Distributed windings



Graphics: Leroy-Somer, Tarun, Nissan, Daimler

Advantages

- Short winding head
- Low production cost
- Typically have higher copper fill factor

- Sinusoidal air gap field (fewer harmonics)
- Usable in most machine types

Disadvantages

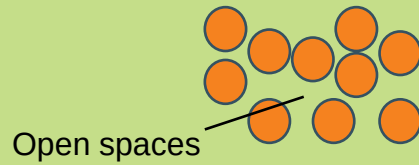
- Many harmonics in the air gap field, torque ripple, $\cos\phi \downarrow$
- Higher rotor losses in asynchronous machines (non-uniform MMF-distribution)

- Large winding head
- Difficult to manufacture

Various technologies with specific advantages and disadvantages with regard to the fill factor are used to manufacture windings.

Characterization of windings according to their structure

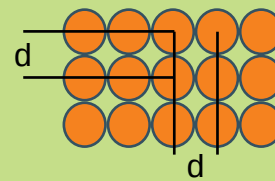
Wild winding



Max. Packing density:

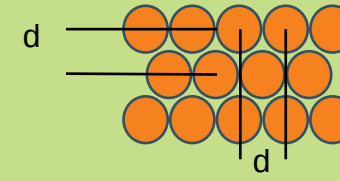
65-75%

Linear winding



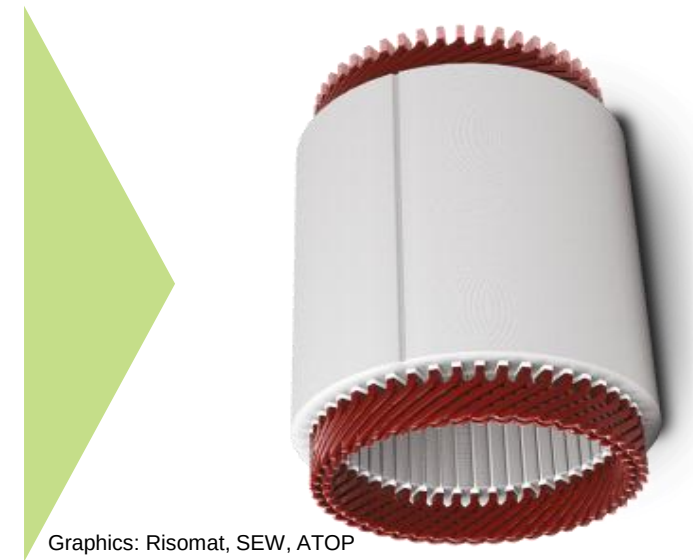
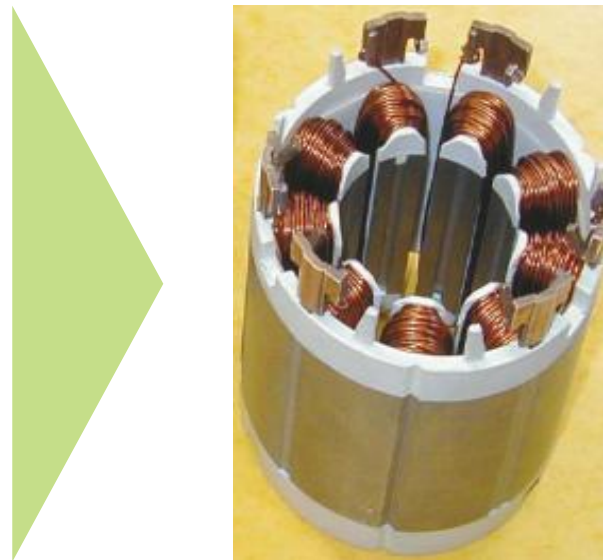
78,5%

Orthocyclic winding



90,7%

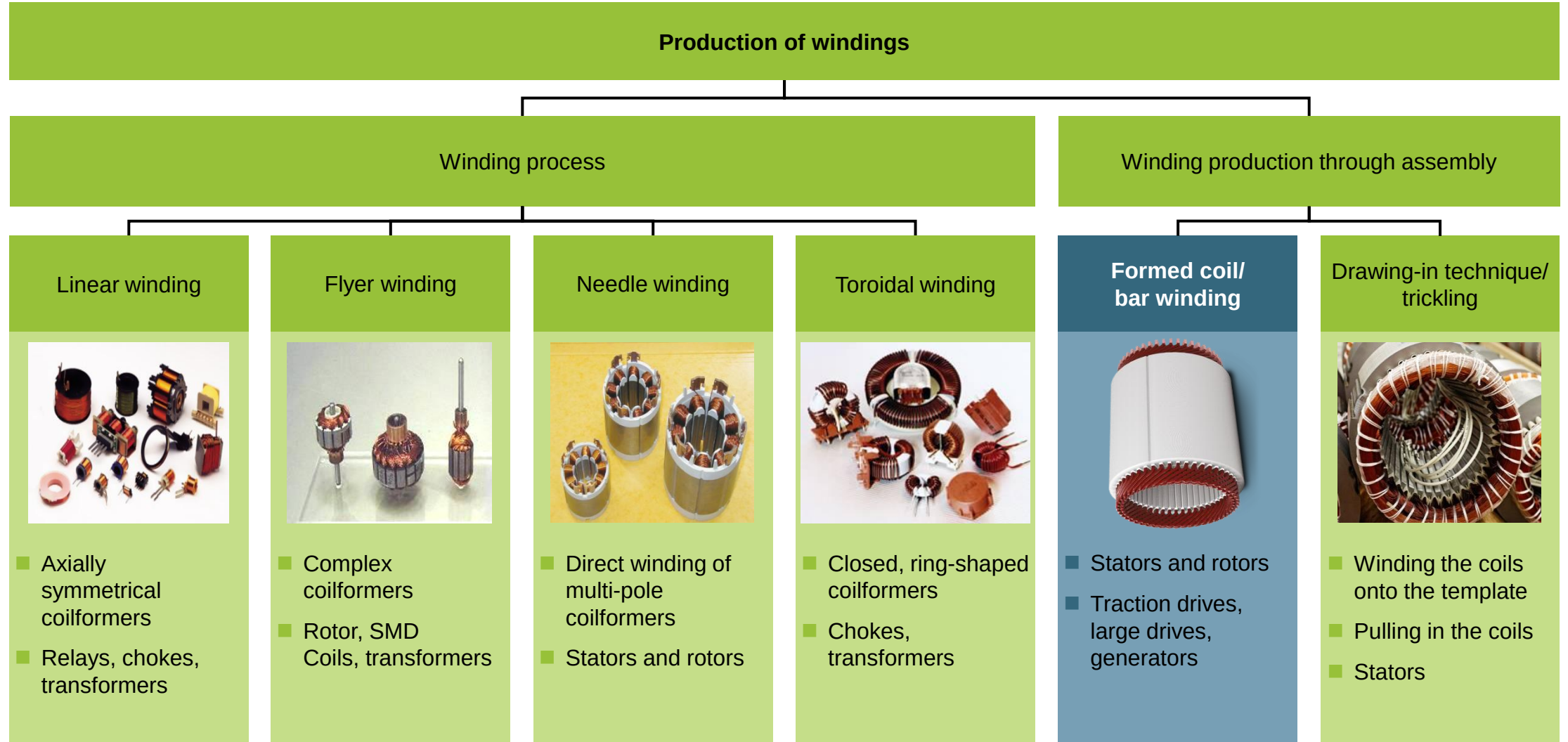
Flat conductor-based
formed coil winding



Graphics: Risomat, SEW, ATOP

There are basically six process variants available for the automated production of windings.

Slide from Lec. 5



Graphics: Kaschke, SEW, FAPS, Westdeutsche Dochtfabrik, ATOP

To meet the requirements of the automotive industry, formed bar and coil technology offers an alternative to conventional round wire winding.

Higher-level requirements

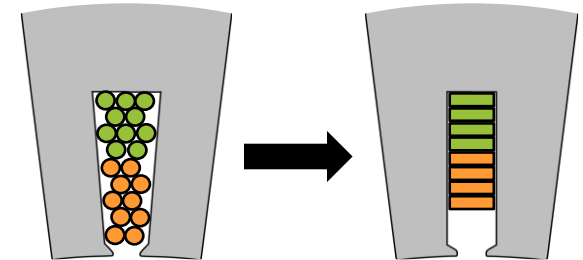
- Increase in power density
- Reduction of installation space and mass
- Fully automated production at low cost



Graphic: Porsche

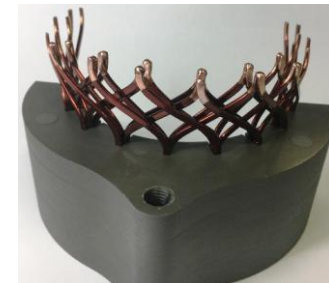
Design aspects of flat conductor-based formed bar or coil windings

- + Increased copper fill factor
- + Better heat dissipation as a result of optimized thermal connection
- + High current-carrying capacity of the individual conductors
- Disadvantages with regard to electricity displacement effects



Advantages and disadvantages in the production of flat conductor-based formed coil and bar windings

- + Simplified production
- + Better automation capability
- + Defined position of the individual conductors in the stator
- High number of contact points



The technologies of formed bar and coil winding have their origins in the manufacture of powerful drives and generators.

Stator wind power generator



Large drive: stator



Large drive: slip ring rotor

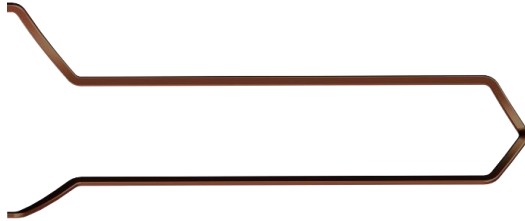
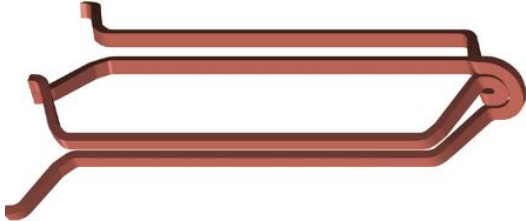
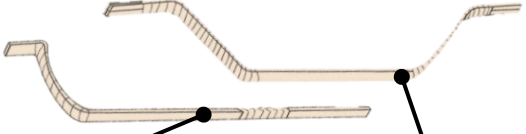
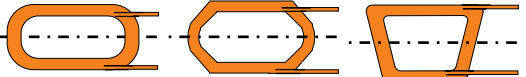


Graphics: Partzsch

Characteristics of the use of formed bars and coils in large electrical machines

- Structure of the windings made of rectangular bare copper wire
- Use of banding (e.g. mica foil) or wrapping (e.g. glass silk) for electrical insulation and fixing (paint insulation not possible due to high voltages)
- Usually open slots to enable installation
- Production of small quantities characterized by a large number of manual activities

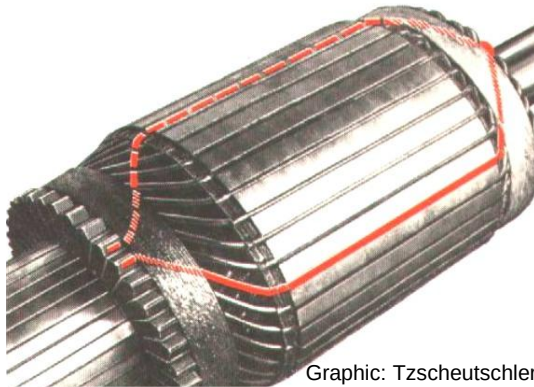
The complexity of open formed coils, closed formed coils and formed bars differs significantly.

- Coil fully formed and insulated outside the machine - No significant deformation when assembled in slot - Differentiation by shape - varying complexity in the production of windings			
	Shape bar	Open formed coil (e.g. hairpin)	Closed formed coil
Geometry			
Forms of execution	 Semi-formed bar Formed bar	--	 Needle coil Oval coil Fishbone
Slot insulation	Sheet metal package / forming coil	Sheet metal package / forming coil	Formed coil
Mounting direction	axial / radial	axial / radial	radial
Design complexity			
Assembly complexity			
Number of contact points			

Loop and wave windings are produced by means of formed bar and coil technology, whereby Roebel bars can be used.

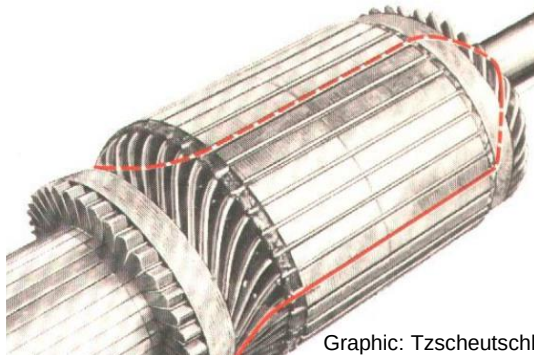
Realizable winding types

Loop winding



Graphic: Tzscheutschler

Wave winding



Graphic: Tzscheutschler

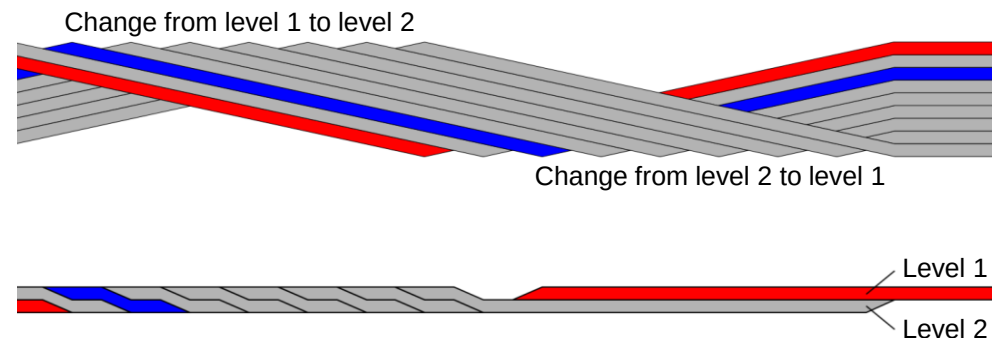
Increasing the efficiency of large electrical machines with Roebel bars

Increasing winding losses at higher frequencies

- Magnetic slot leakage field or transverse field causes current to be diverted to the slot head
- Uneven current density in winding parts causes additional losses

Use of Roebel bars to reduce current displacement losses

- Dividing the conductor into several insulated partial conductors
- Changing the vertical position of the partial conductors in the slot to achieve a uniform current density in the partial conductors



Graphic: Christian Barth



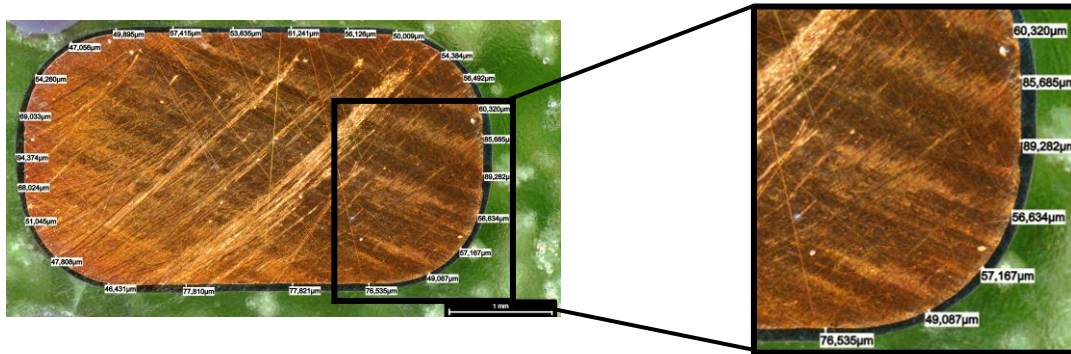
Graphic: Partzsch

Significant increase in the manufacturing costs of the winding

Formed coils and bars for use in automotive traction machines are often made from rectangular enameled insulated copper wire supplied on spools.

Structure of the rectangular enameled insulated conductor

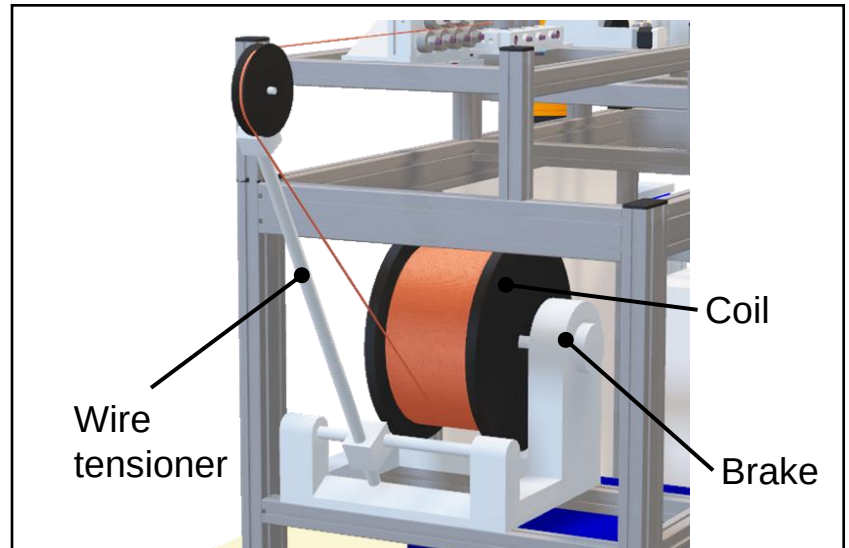
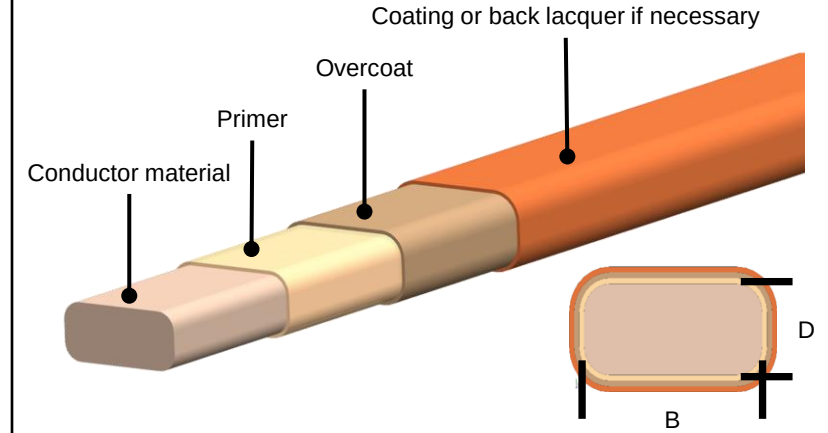
- Flat conductors available in different heights and widths ($h : t = 8 : 1$ to $1.4 : 1$)
- Multi-layer structure
- Classification by heat class (e.g. $H = 180\text{ }^{\circ}\text{C}$)
- Differentiation according to increase in cross-section due to insulation (grade 1 to grade 3)
- Inhomogeneous distribution of insulation layer thickness



Supply of the enameled copper wire

- Provision of the conductors as a wire coil on a spool
- Braking of the spool ensures constant unwinding movement
- Feeding the enameled copper wire to further production steps at a defined position

Structure of enamelled insulated conductors



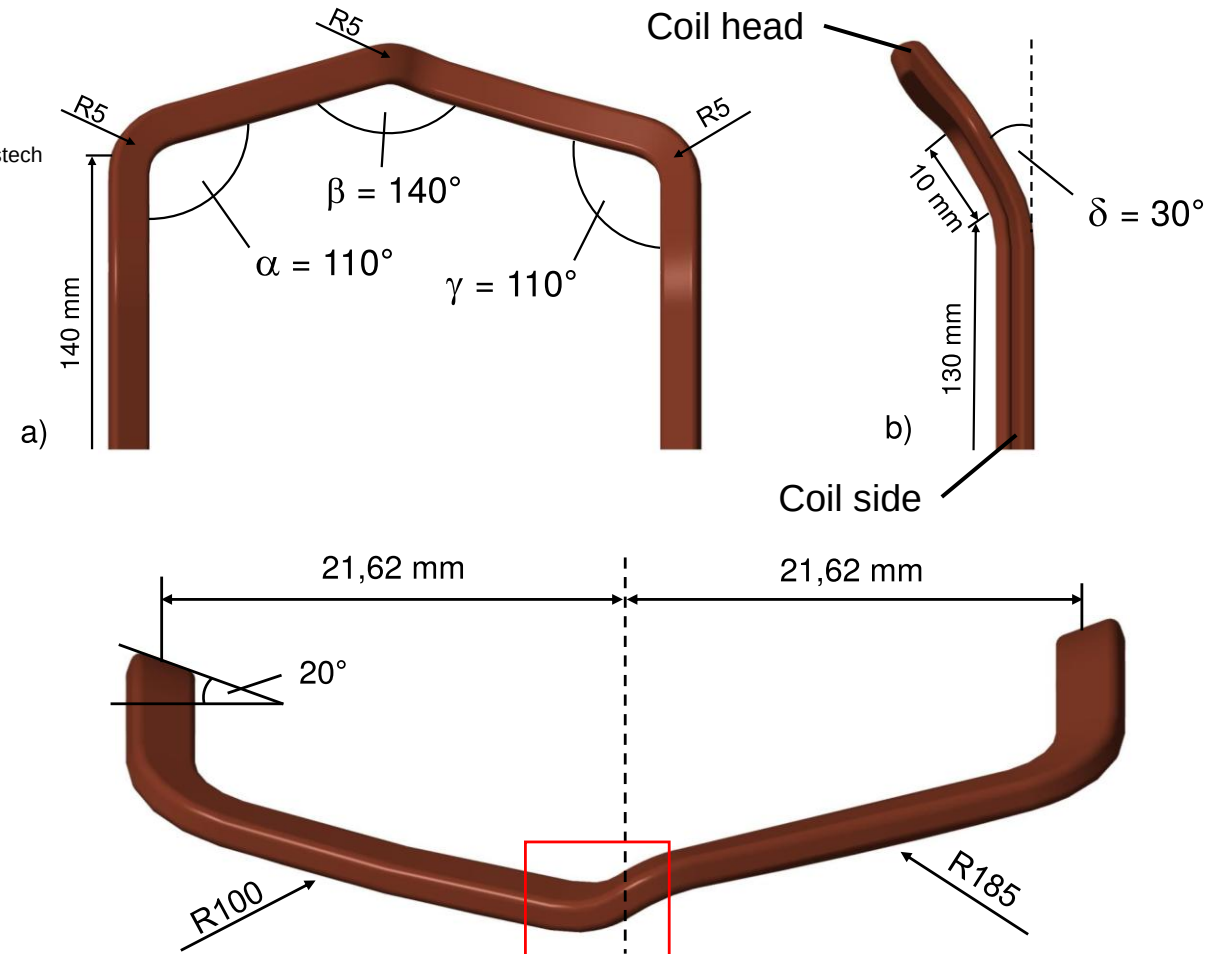
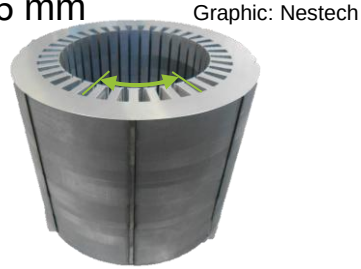
An open formed coil realized in laboratory operation illustrates the complexity of the head geometry.

Wire

- "Thermex® 220 flat"
- Conductor cross-section 3.15 mm x 1.6 mm

Design guidelines

- The width of the slot step decreases in the direction of the inside diameter
- Formed coils must have have varying widths depending on their radial position
- The number of variants of forming coils must be minimized
- 3D contour in the coil head to avoid collisions
- Twist the coil side so that it runs parallel to the slot base
- Maintain bend radii between 3 mm and 5 mm
- Grade 3 insulation is often used to prevent short circuits



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Graphic: ATOP

The production chain for the manufacture of a stator with a distributed winding of round wire essentially consists of nine steps.

Sheet metal package

- Sheet metal cut prod.
- Stacking sheet cuts
- Joining sheets



Insulation slot base

- Folding insul. paper
- Insert paper
- Cut off



Winding stator

- Direct winding production
- Indirect winding production



Inserting the cover slide

Closing open slots with insulating closure



Switching and contacting

- Bundling the phases
- Contacting



Fixing the winding head

- Isolating the phases
- Bandaging wrapping head



Pressing winding head

Reduced space requirement for winding head



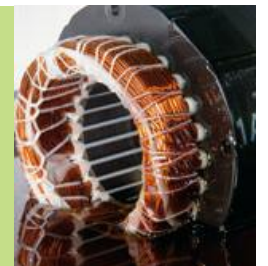
Electrical testing

- Current voltage test
- Measurement of winding resistance

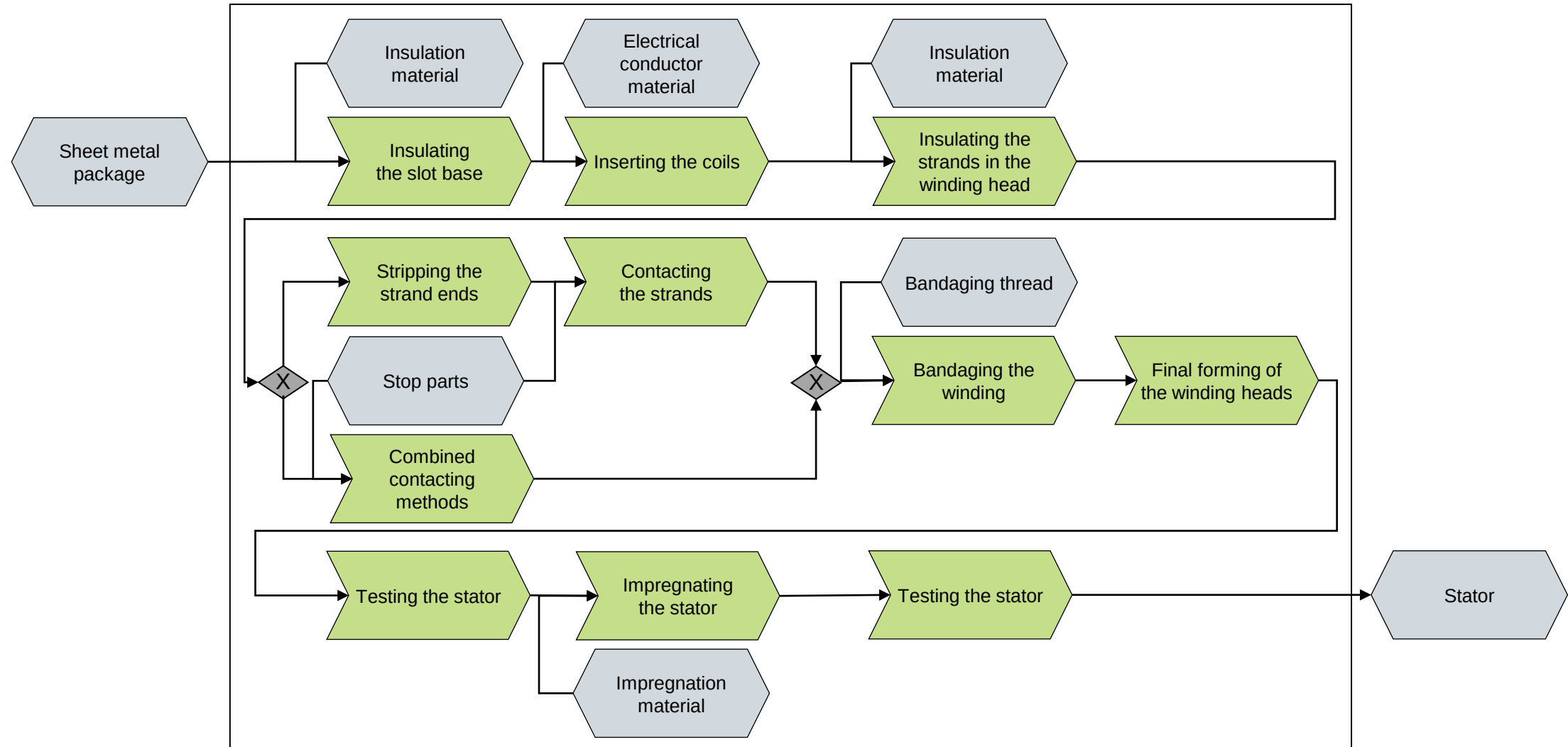


Impregnation

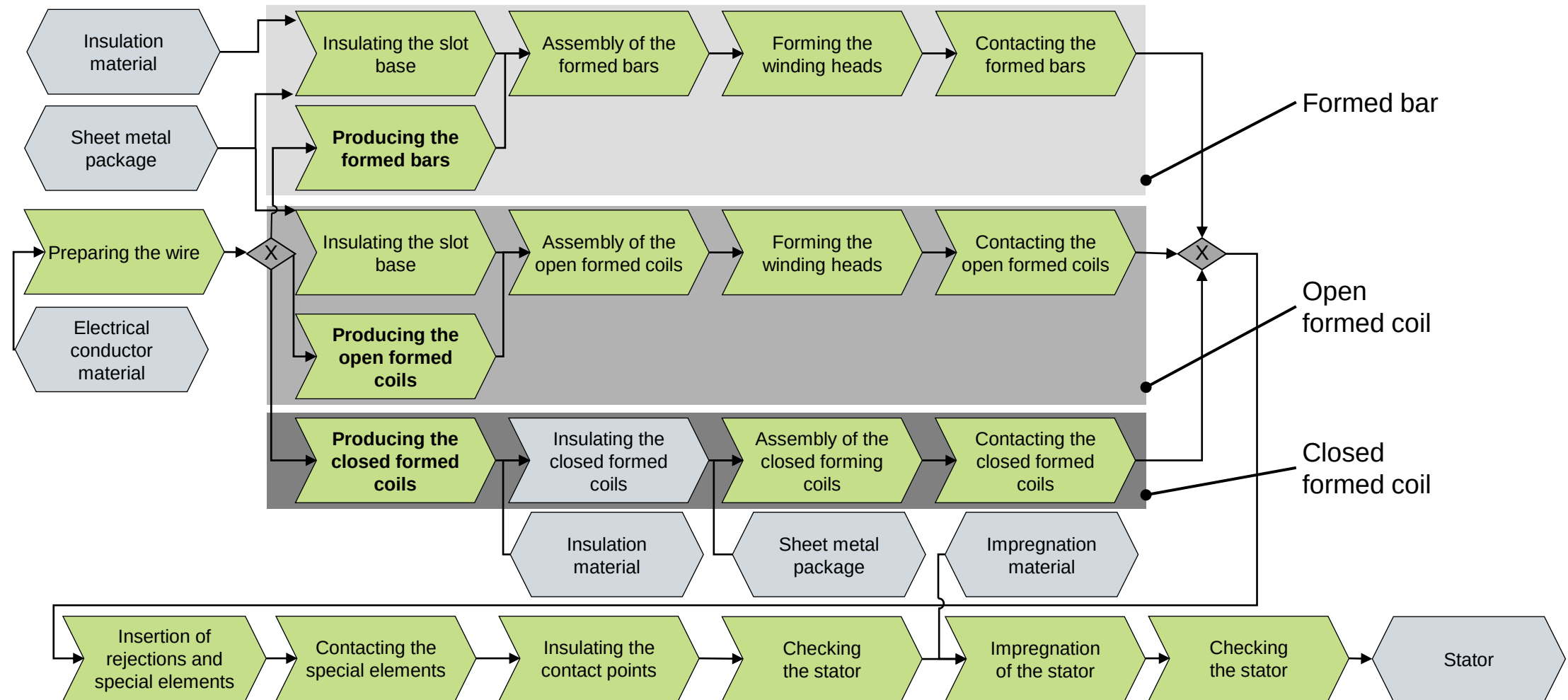
- Application of impregnation material
- Curing impregnation material



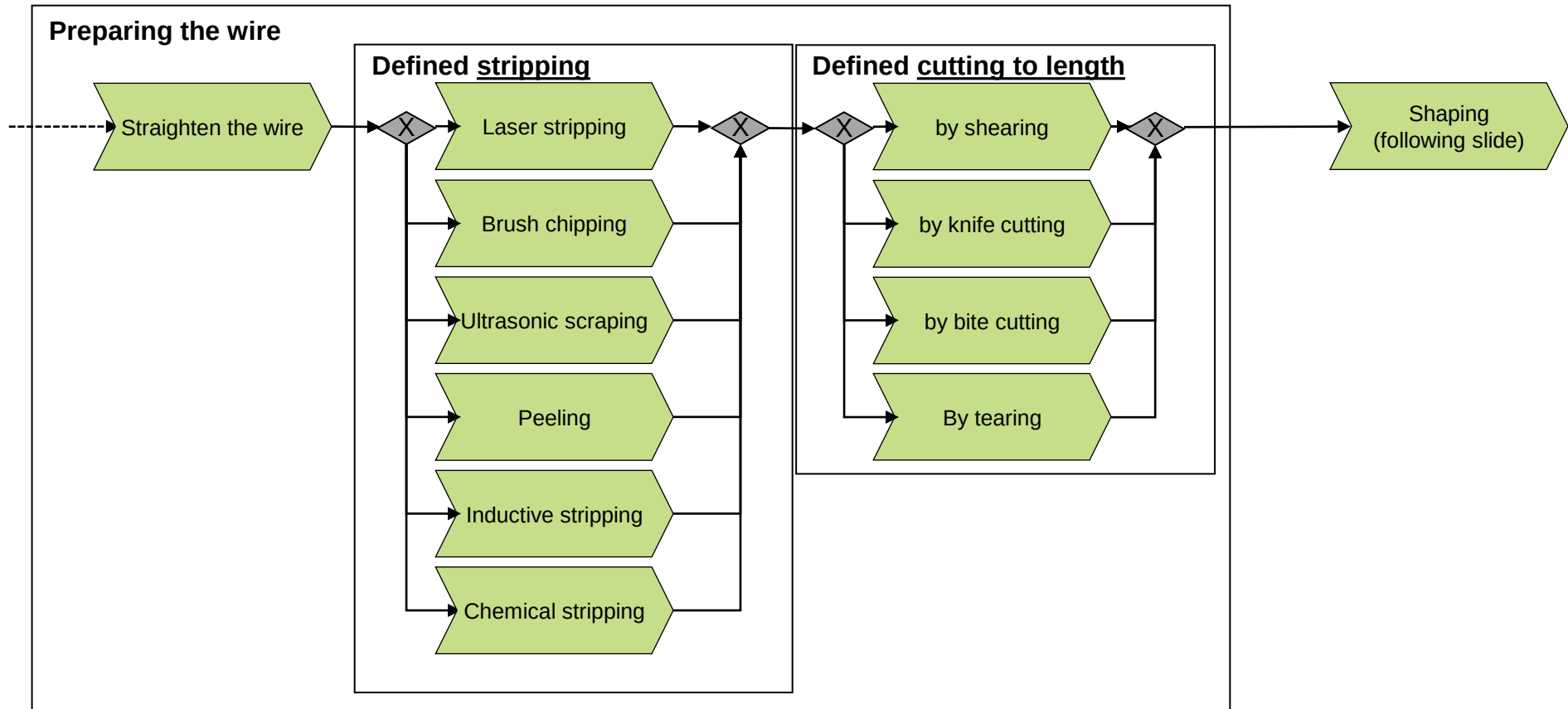
The production chain for the manufacture of a stator with a distributed winding of round wire essentially consists of nine steps.



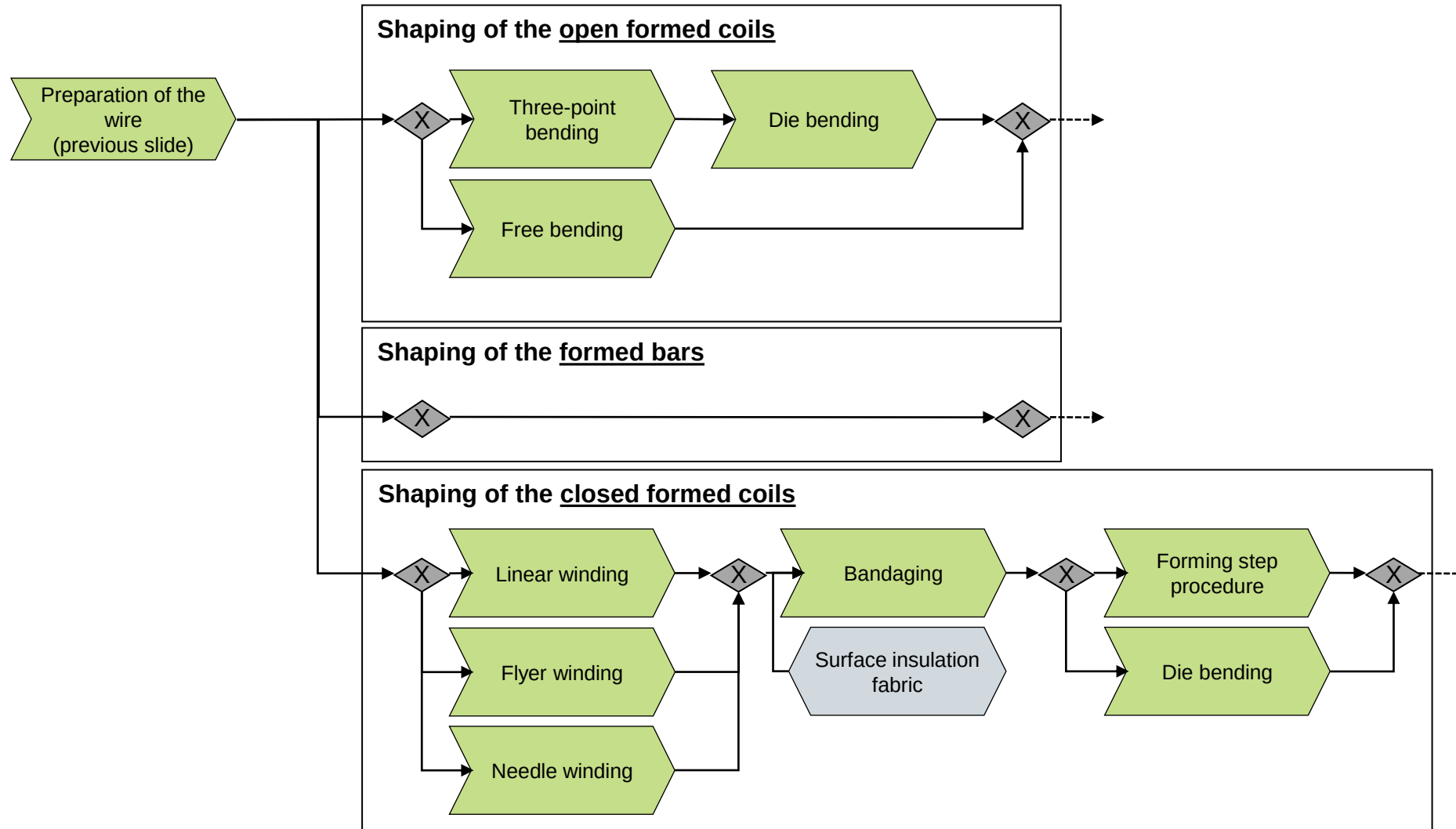
The process chains differ significantly depending on whether formed bars, open formed coils or closed formed coils are used.



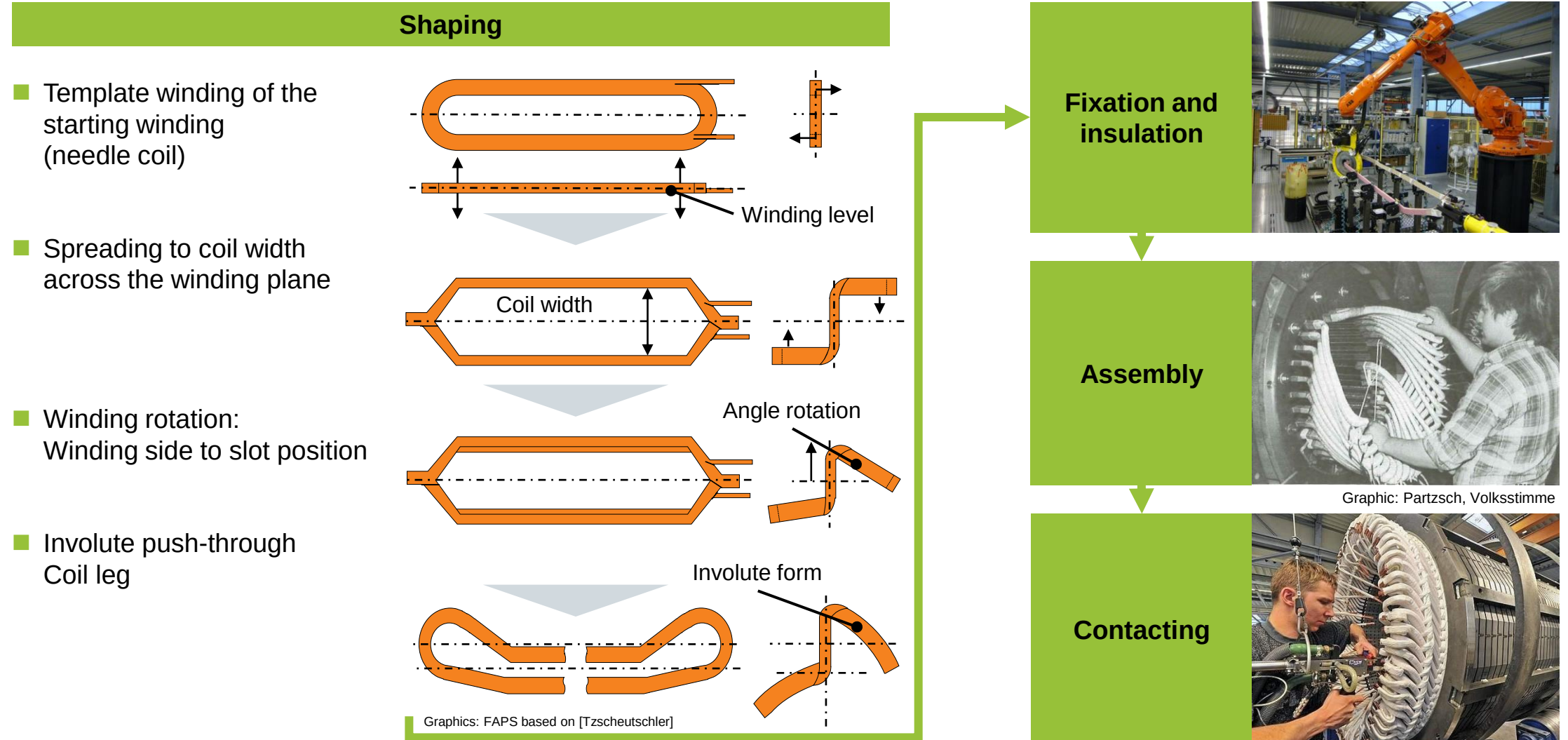
There is no difference in the preparation of the wire for the production of formed bars and open/closed formed coils.



There is no difference in the preparation of the wire for the production of formed bars and open/closed formed coils.



The manufacture of electrical machines with closed formed coils is characterized by complex shaping.



When using closed formed coils, shaping and assembly are challenging, but the number of contact points to be made is reduced.

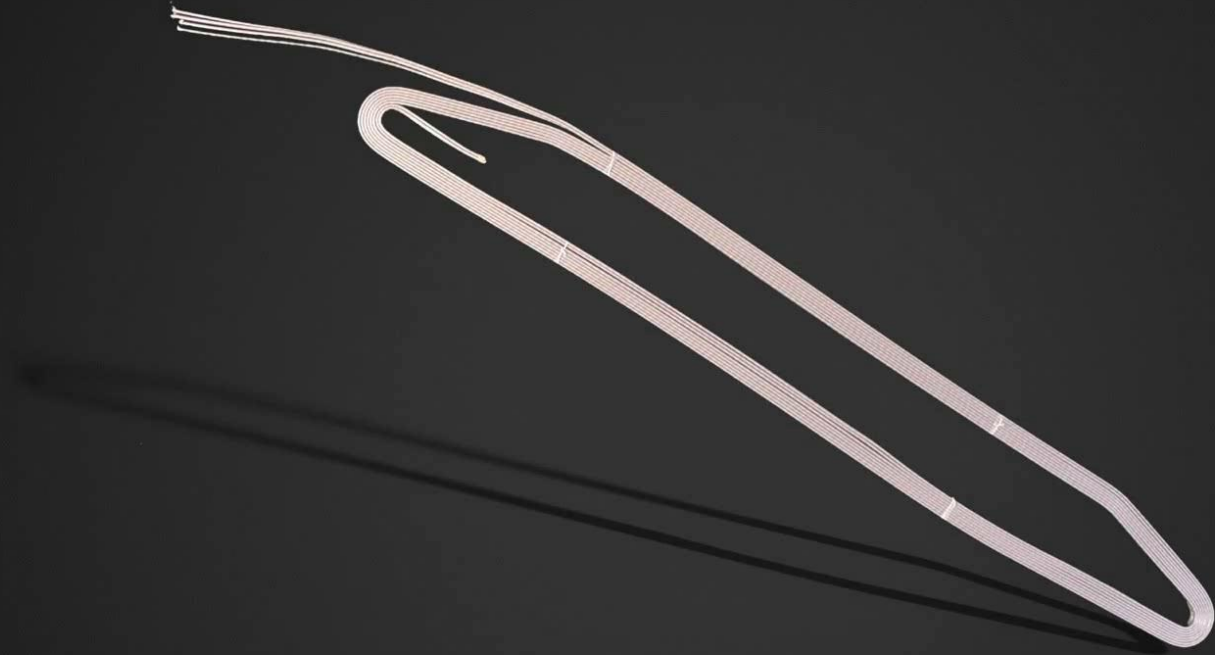
Process

Process chain for the production of
stators with closed formed coil
windings

Video:
J. Helmke & Co.

[youtube.com/watch?
v=tEE7TcJRGY8](https://youtube.com/watch?v=tEE7TcJRGY8)





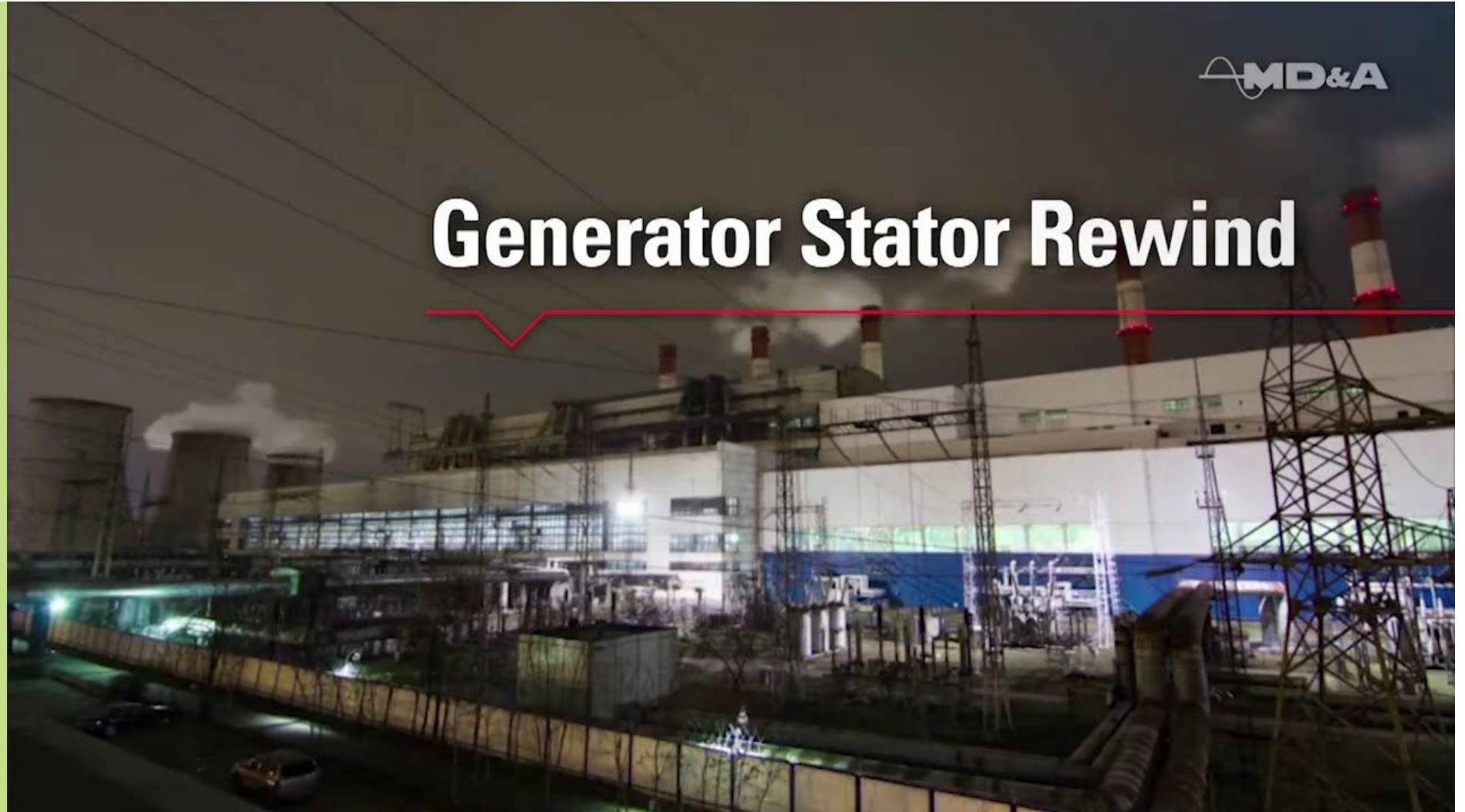
The manufacture of large electrical machines is characterised by manual activities.

Process

Reworking of the formed bar
winding of a generator

Video:
MD&A Turbines

[youtube.com/watch?
v=hnne3Sm-5jU](https://youtube.com/watch?v=hnne3Sm-5jU)



Open formed coils made of rectangular enameled copper wire are suitable for axial mounting and are increasingly being used for smaller generators/alternators & traction drives.



Use of open formed coils in medium and low power electrical machines

- Use of enameled insulated copper conductors
- Axial mounting enabled by insertion
 - Direct insertion of slot insulation in sheet metal package
 - Assembly of open formed coils even with closed slots
- Necking and twisting required for correct positioning of the conductor ends after installation and before contacting
 - If necessary, radial bending of the conductor ends („necking”) to create space between levels
 - Cabinets for joining conductor ends to be connected („twisting”)

Video:
GM

[youtube.com/watch?v=8bbjb7_jX6c](https://www.youtube.com/watch?v=8bbjb7_jX6c)



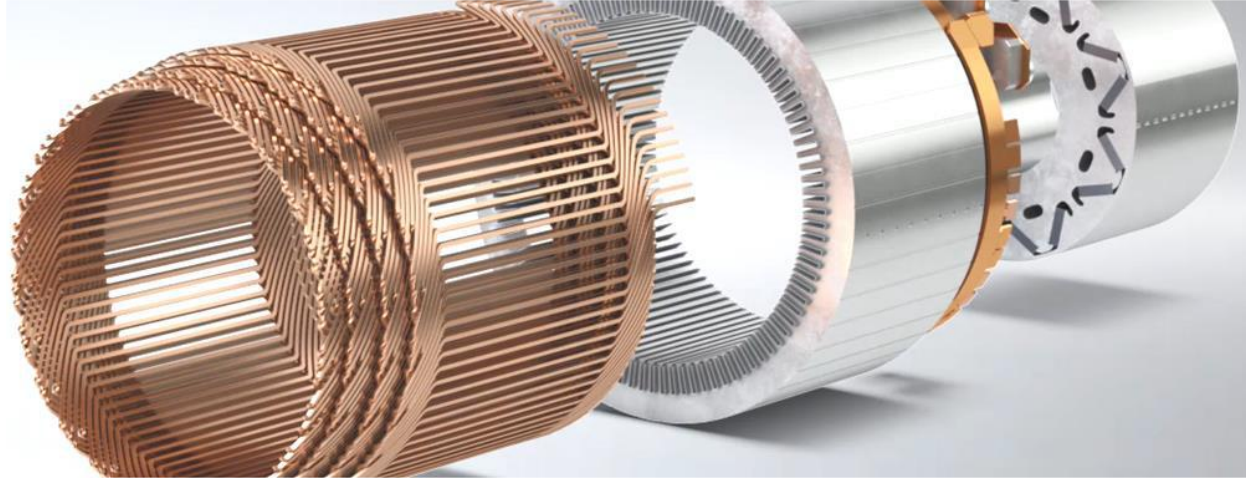
Open formed coils can be manufactured using free-form bending or tool-based processes.



The production and assembly of formed bars is carried out in simple processes, but a high number of contact points must be realized.



Continuous hairpin winding is a special case of open formed coil winding, which has the advantage of reduced contacting costs.



Continuous hairpin winding

- Winding strands are produced continuously without interruptions
- + Elimination of contacting operations
- + Higher number of windings possible
- + Use of flatter conductors possible
- Installation only possible with open slots

Graphics: ELMOTEC STATOMAT

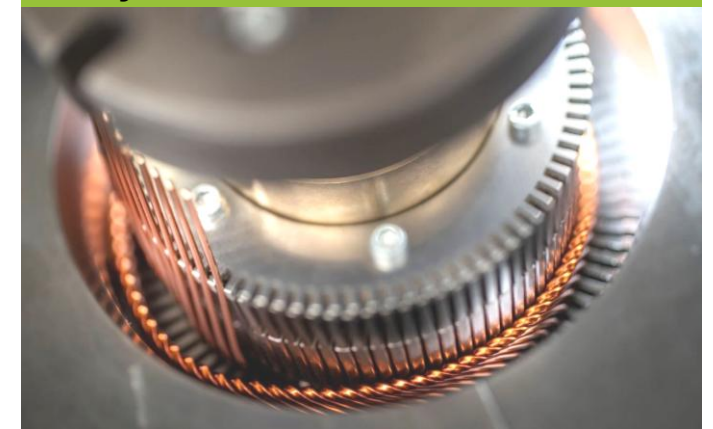
Forming the winding in the plane



Rolling up the winding mat



Assembly of the winding by insertion in radial direction



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Graphic: ATOP

To avoid short circuits, the slot base is insulated in different ways when manufacturing stators with open formed coil windings.

Task

- Electrical separation of wire and sheet metal package
- Protection of the enameled wires from punching edges and possible damage

Requirements

- Automated positioning
- Small dimensions to not limit available winding space

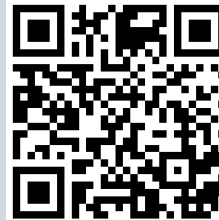
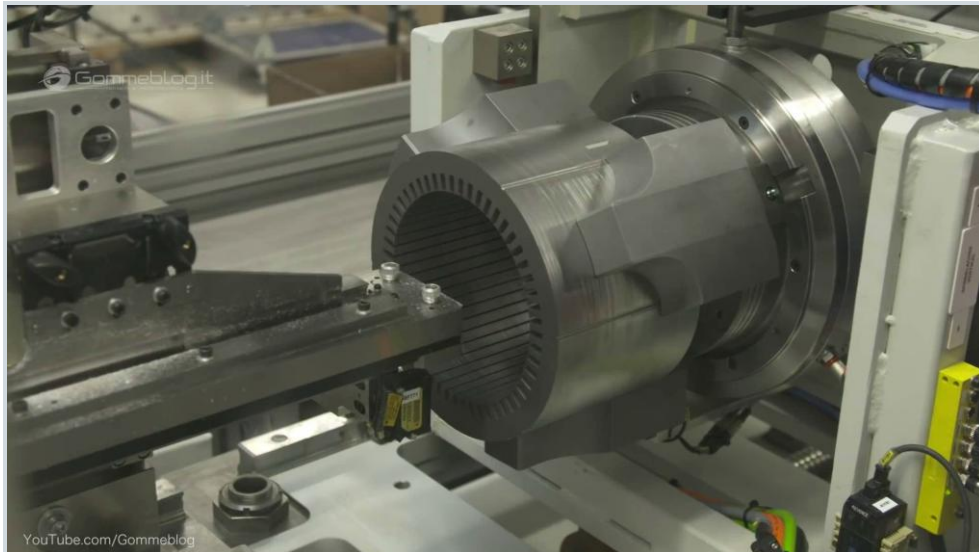
Procedures

Plastic films

Powder coating

Slot liner

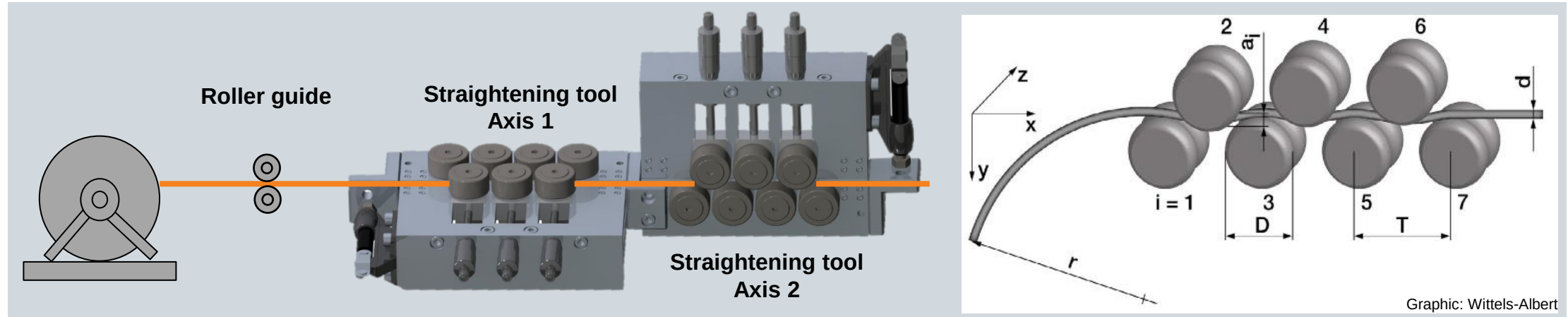
KS formed parts



Video:
gommelblog.it
[youtube.com/watch?v=xvaQMTcckSg](https://www.youtube.com/watch?v=xvaQMTcckSg)

- Production of insulation layers by means of cut-to-length and formed layer laminates
- Established technology
- Materials:
 - Mica paper
 - Aramid paper (Nomex ®)

The straightening process is the basis for the realization of dimensionally stable flat conductor-based formed coils from enameled copper wire.



Description

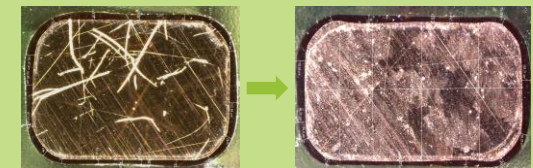
- Reduction of residual stresses in semi-finished products
- Compensation of bends in the wire after unwinding from the coil
- Alignment determines achievable process results of the following steps

Process

- Straightening tool consisting of several rolls
- Distance between rollers are precisely adjustable
- Repeated elastic-plastic deformation in opposite directions
- Parameters
 - Position of straightening rollers
 - Feed rate

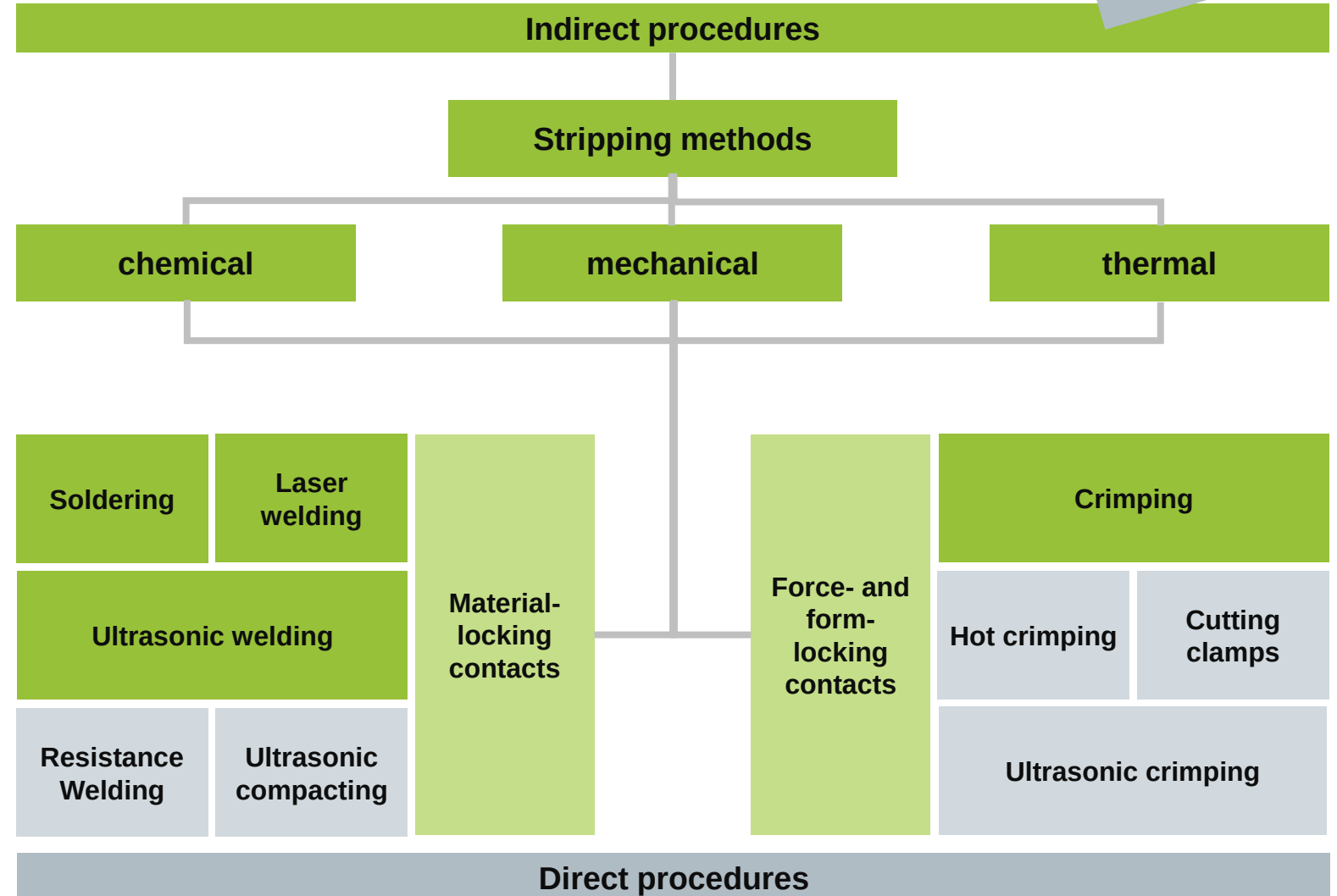
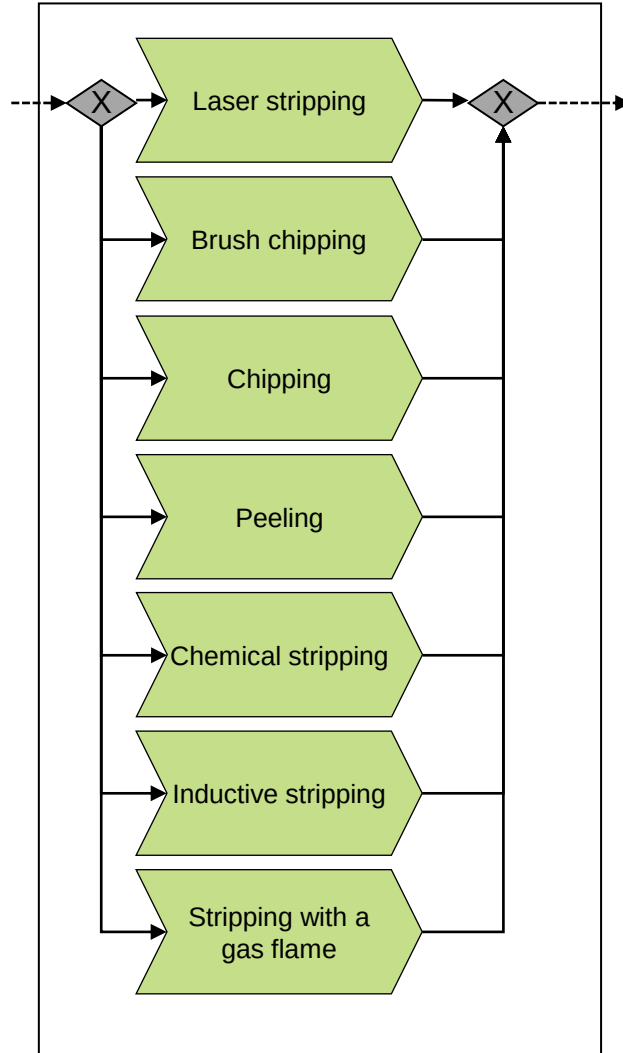
Challenges

- Avoids damage to the primary insulation
- Fluctuating cross-sections of the starting material
- Plastic deformation of the conductor cross-section is not permitted



In indirect contacting processes, the enameled copper wires are first stripped, whereby the stripping methods can be divided into three main groups.

Slide from Lec. 7



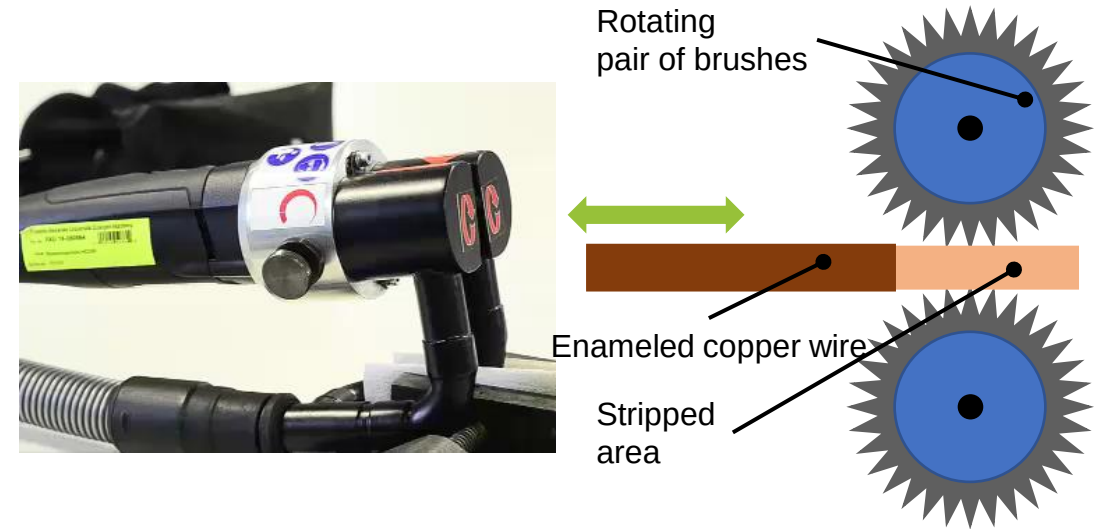
The insulation coating can be removed in different ways during the automated production of formed coils and bar windings.

Slide from Lec. 7

Rotating brushes

The enameled copper wire is inserted between two opposing rotating steel brushes, whereby the insulation layer is removed and vacuumed off.

- Rotating steel brushes are arranged opposite each other
- Distance is adapted to the wire diameter
- Inserting the insulated copper wire,
- Removing the insulation due to friction and



Advantages

- Low costs
- Simple process control
- Easy to use
- Suitable for various cross-sections
- Applicable for small quantities
- State of the art

Disadvantages

- Fluctuating removal quality and tool wear
- Damage to the wire due to mechanical forces
- Danger of insulation residues
- Undefined transition between stripped and insulated area
- Damage to the conductor surface by brush wires
- Not suitable for melting thermoplastics

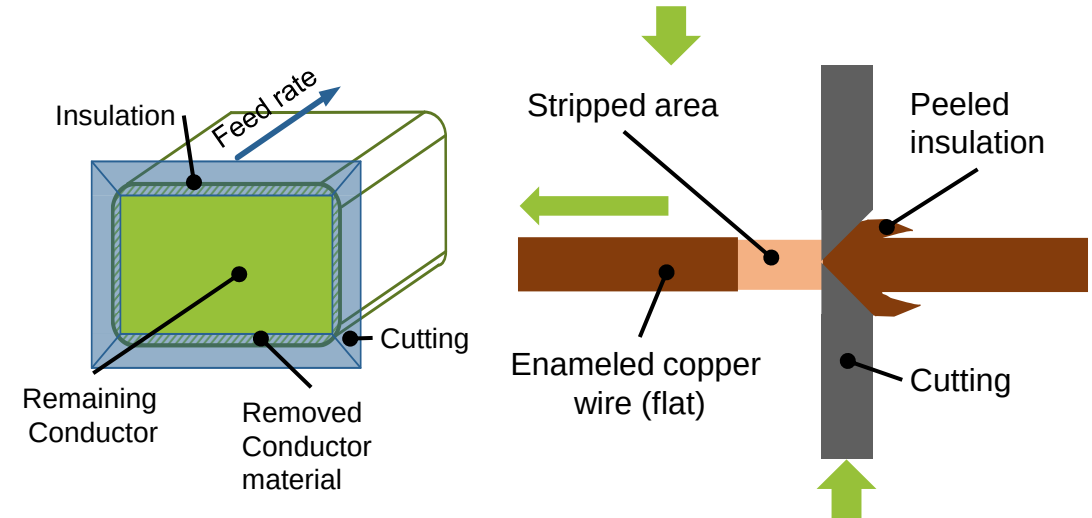
The insulation coating can be removed in different ways during the automated production of formed coils and bar windings.

Slide from Lec. 7

Peeling

Blades cut into the copper material in a rectangular geometry and realize a feed movement along the conductor, whereby the insulation is removed without leaving any residue.

- Kinematic structure usually consisting of four cutting edges
- Cutting edges penetrate the insulation and cut into the conductor material, linear feed movement of the cutting edges along the conductor or of the conductor along the cutting edges leads to peeling of the insulation
- Due to the shape of the flat copper conductors (radii at edges), removal of conductor material cannot be avoided



Advantages

- Easy to use
- Defined cutting line
- Defined transition area
- Simple process principle
- Independent of insulation type
- residue-free removal

Disadvantages

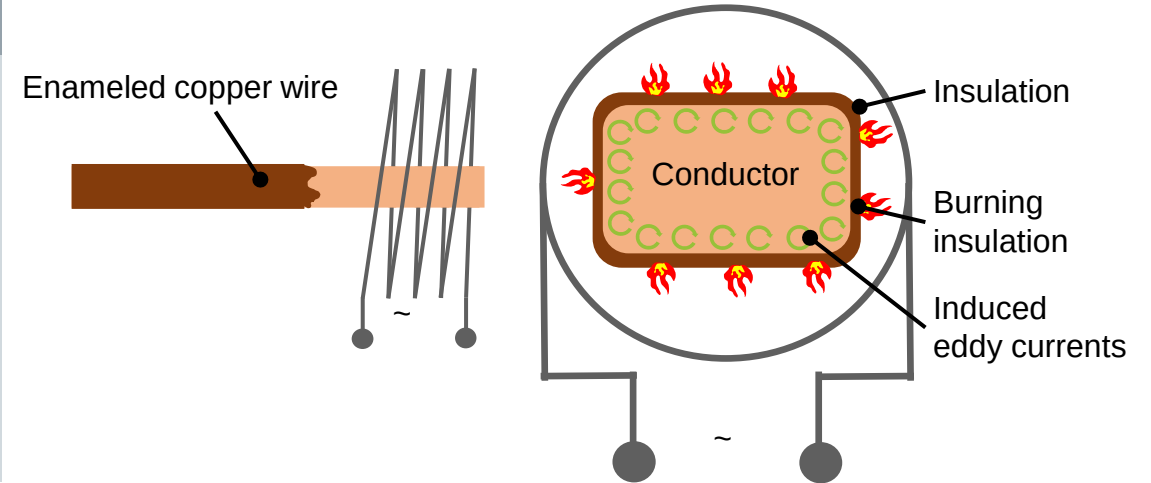
- Removal of conductor material by penetration of the cutting edges into the copper material
- Depth must be adapted to the edge radius
- Tool wear
- Challenging positioning of the cutting edges

The insulation coating can be removed in different ways during the automated production of formed coils and bar windings.

Slide from Lec. 7

Inductive heating

The conductor is placed in the center of a coil that is connected to a high-frequency alternating voltage. The resulting alternating magnetic field induces eddy currents. The resulting high current density on the surface leads to severe heating of the conductor and burns the insulation.



Advantages

- Short process times
- Non-contact method

Disadvantages

- Oxidation surface
- Heat input damages insulation outside the area to be removed
- Combustion residues

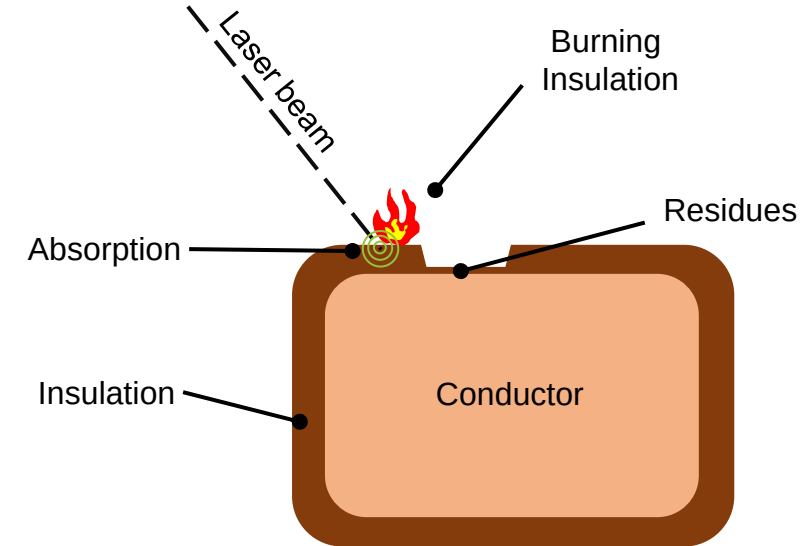
The insulation coating can be removed in different ways during the automated production of formed coils and bar windings.

Slide from Lec. 7

Laser: Long-wave infrared radiation

Long-wave infrared radiation is reflected by the copper conductor, while it is absorbed in insulation material. This leads to burning of the insulation.

- Use of CO₂ lasers ($\lambda \sim 10,600$ nm, pulsed)
- Burning the insulation material
- No interaction with the conductor
- Insulation residues in the size range of the wavelength



Advantages

- Non-contact method
- High removal rate
- Defined transition area

Disadvantages

- Insulation residues in the area of $\lambda/4$
- Complex process
- High investment costs

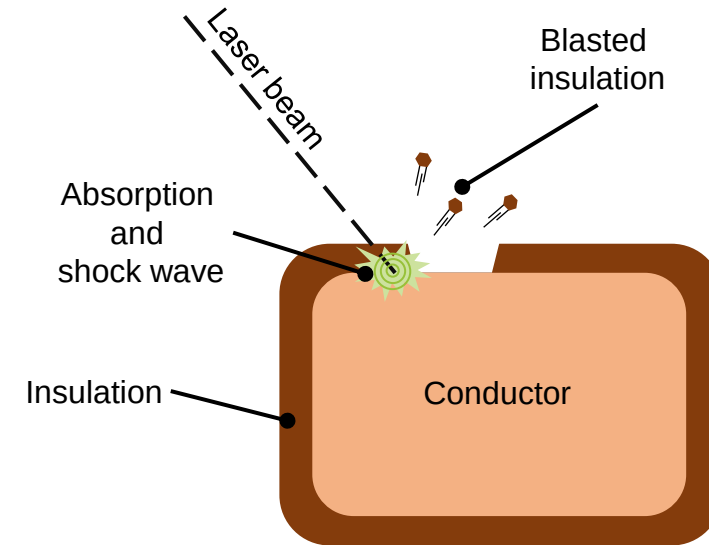
The insulation coating can be removed in different ways during the automated production of formed coils and bar windings.

Slide from Lec. 7

Laser: Near infrared radiation

Near infrared radiation is partially absorbed by the surface of the copper conductor after passing through the insulation. The pulsed laser power causes the copper material to heat up abruptly and expand without melting.

- Use of solid-state lasers (e.g. Nd:YAG laser, $\lambda \sim 1,064 \text{ nm}$) in pulsed operation
- Shock waves are generated which cause the insulation to flake off



Advantages

- Non-contact method
- Free of residues
- Defined cutting line
- High removal rate

Disadvantages

- Complex process
- High investment costs

Short-pulse lasers in the infrared wavelength range can be used for the removal of the insulating coating.

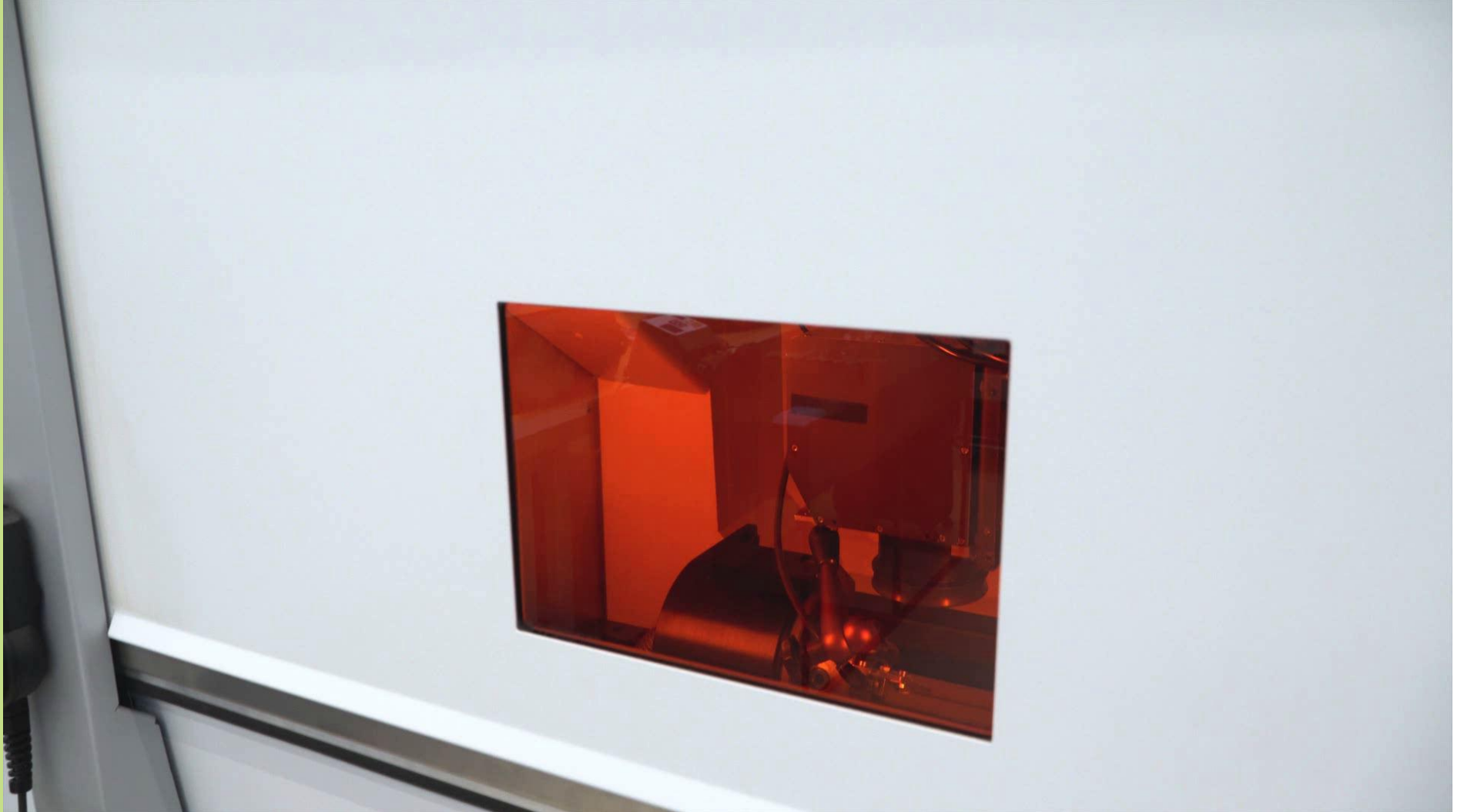
Slide from Lec. 7

Process

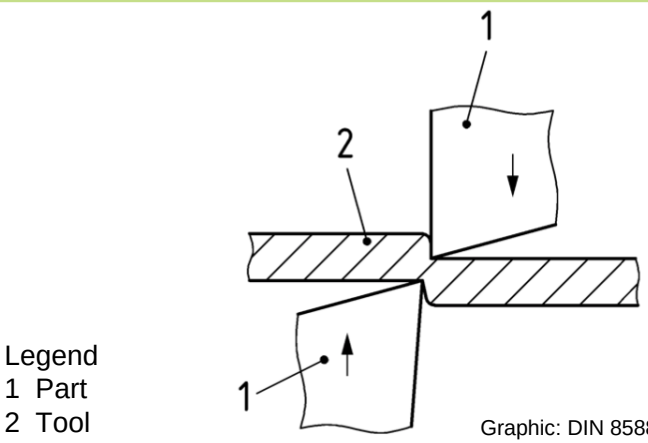
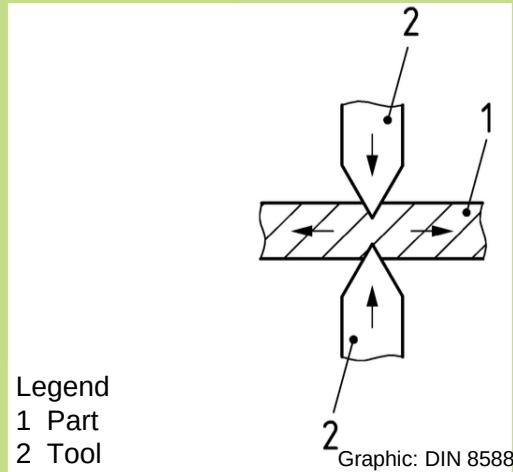
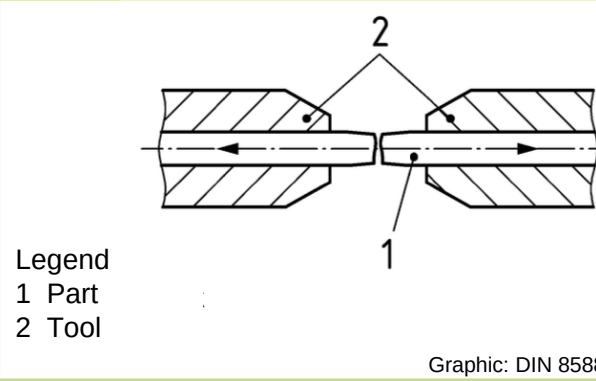
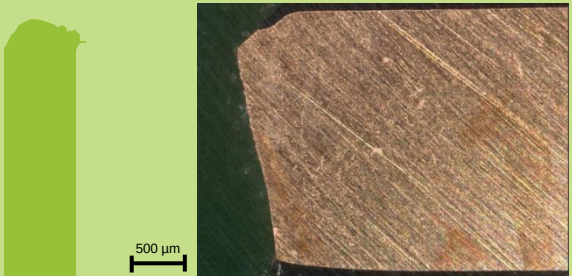
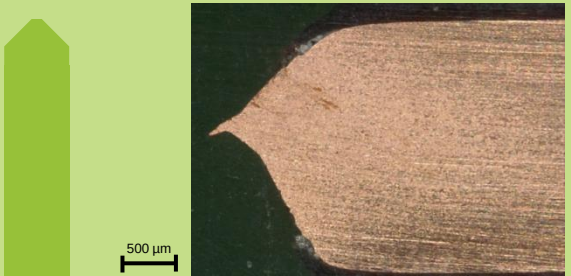
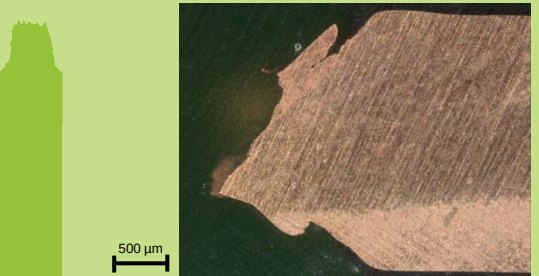
Removal of the paint insulation
with a short-pulse laser

Video:
Trumpf

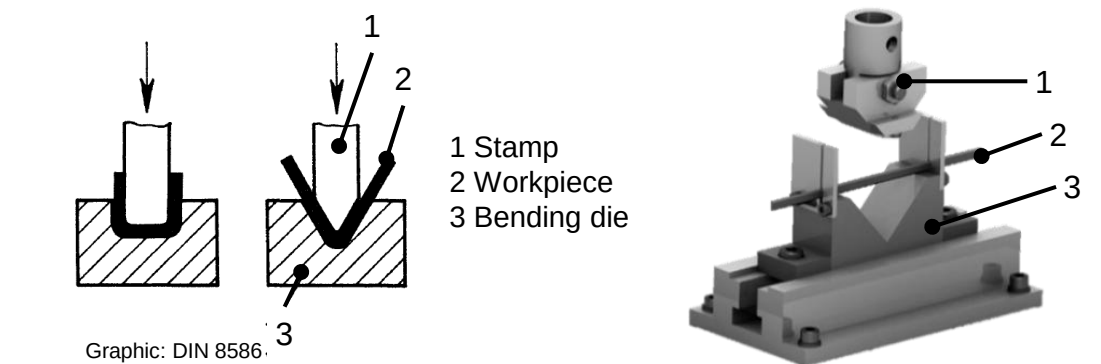
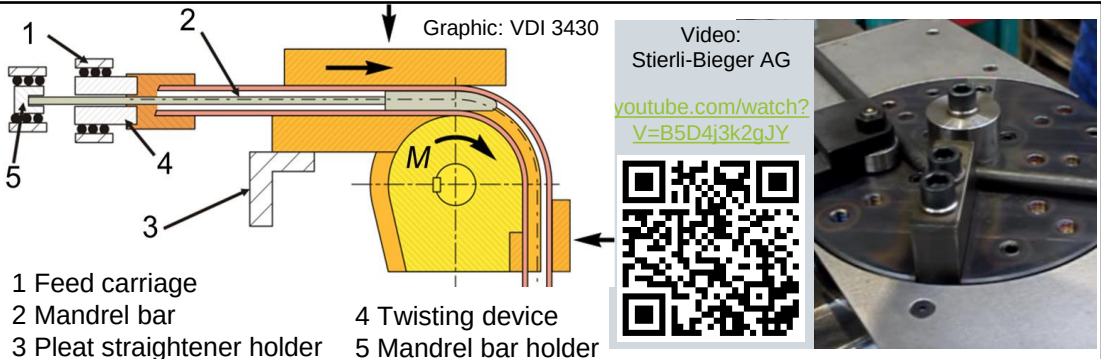
[youtube.com/watch?
v=kW0zPOvUY2M](https://youtube.com/watch?v=kW0zPOvUY2M)



Various technologies described in DIN 8588 (Separation without producing shapeless materials) can be used to separate the conductors after the removal of the coating.

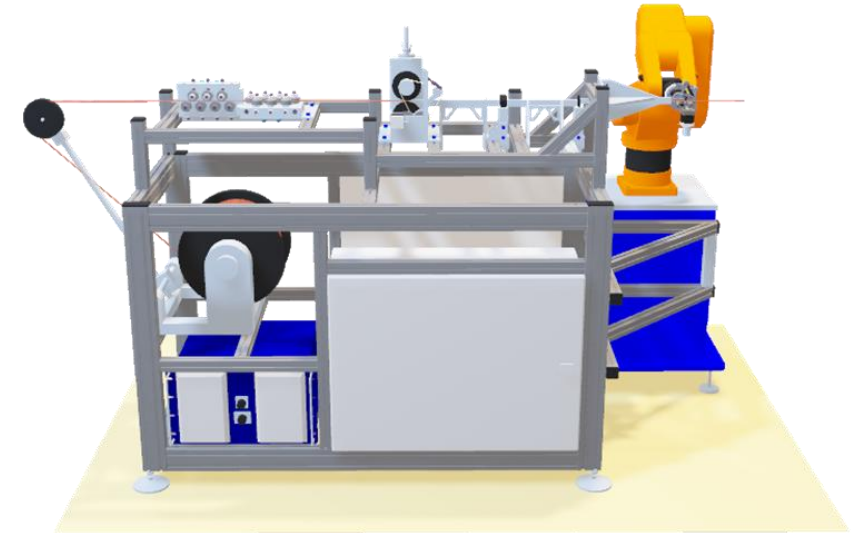
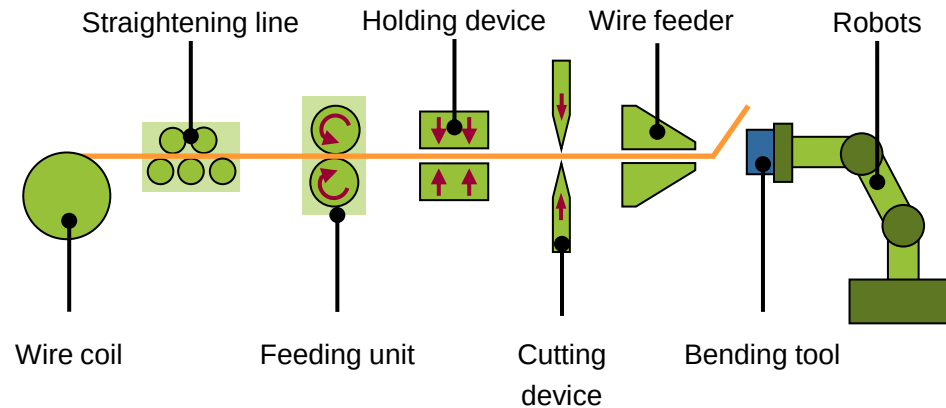
Shear cutting	Bite cutting	Tearing
Single-stroke shear cutting  Legend 1 Part 2 Tool Graphic: DIN 8588	Single-stroke bite cutting  Legend 1 Part 2 Tool Graphic: DIN 8588	Notching (bite cutting) followed by tearing  Legend 1 Part 2 Tool Graphic: DIN 8588
Cut surface with smooth cut and broken surface:  500 μm	Tool-related cut surface:  500 μm	Plastic deformation in pulling direction:  500 μm

In order to realize the geometry of formed coils, die bending and rotary draw bending are common methods for realizing the geometry of formed coils.

Requirements	- Compliance with shape and dimensional tolerances - Low production and cycle times - High reproducibility - No damage to the insulation layer	 Graphic: ATOP
Die bending	Advantages and disadvantages	- Automation capability (++) - Process reliability (+) - Cycle time (++) - Flexibility (--) - Tool costs (-)
Rotary draw bending	Advantages and disadvantages	- Automation capability (+) - Flexibility (+) - Tool costs (+) - Process reliability (o) - Cycle time (-)  1 Stamp 2 Workpiece 3 Bending die Graphic: DIN 8586  Graphic: VDI 3430 Video: Stierli-Bieger AG youtube.com/watch?v=B5D4j3k2gJY 1 Feed carriage 2 Mandrel bar 3 Pleat straightener holder 4 Twisting device 5 Mandrel bar holder

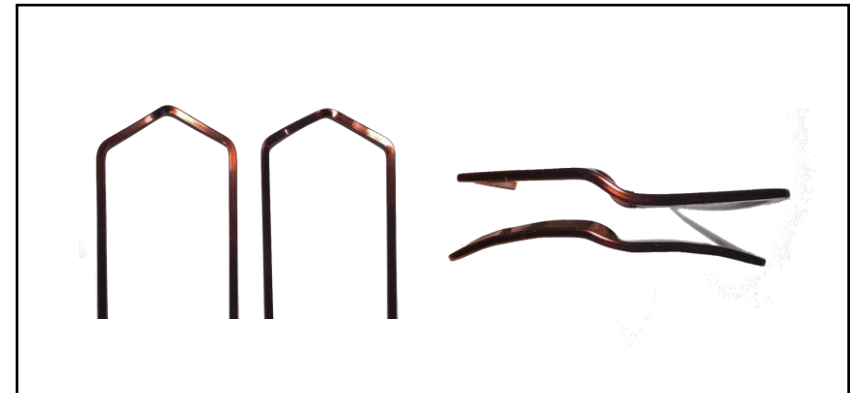
The forming of flat wires using a rotary bending process offers further degrees of flexibility compared to conventional die bending.

System structure



Results

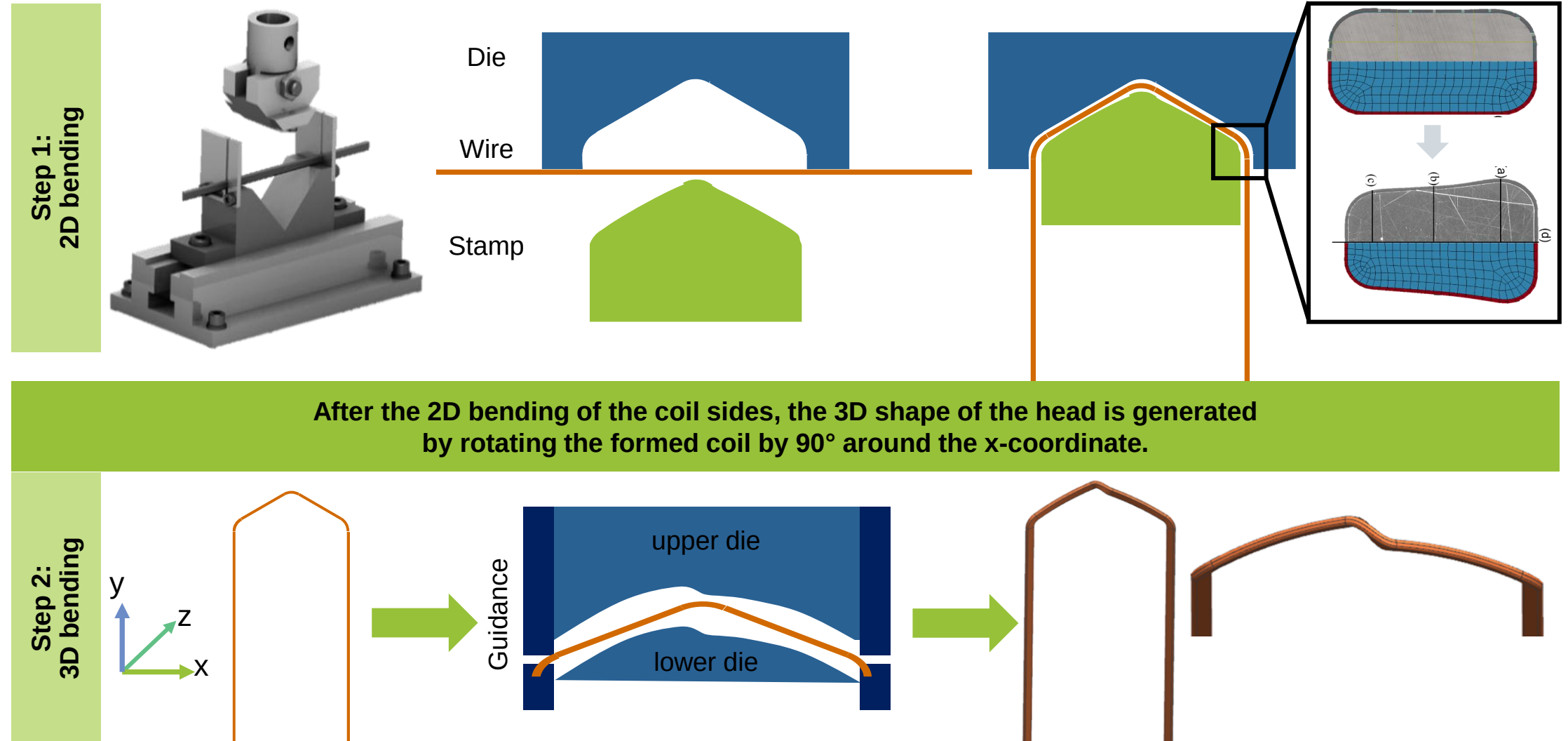
- Rotary tension bending of hairpins implemented and proven
- Error influences reduced by linking the system
- Processing of different wire dimensions possible
- Various hairpin dimensions can be produced
- The influences on the primary insulation during the bending process require further investigation



Rotational tension bending is characterized by its high flexibility and is therefore very suitable for prototyping activities.

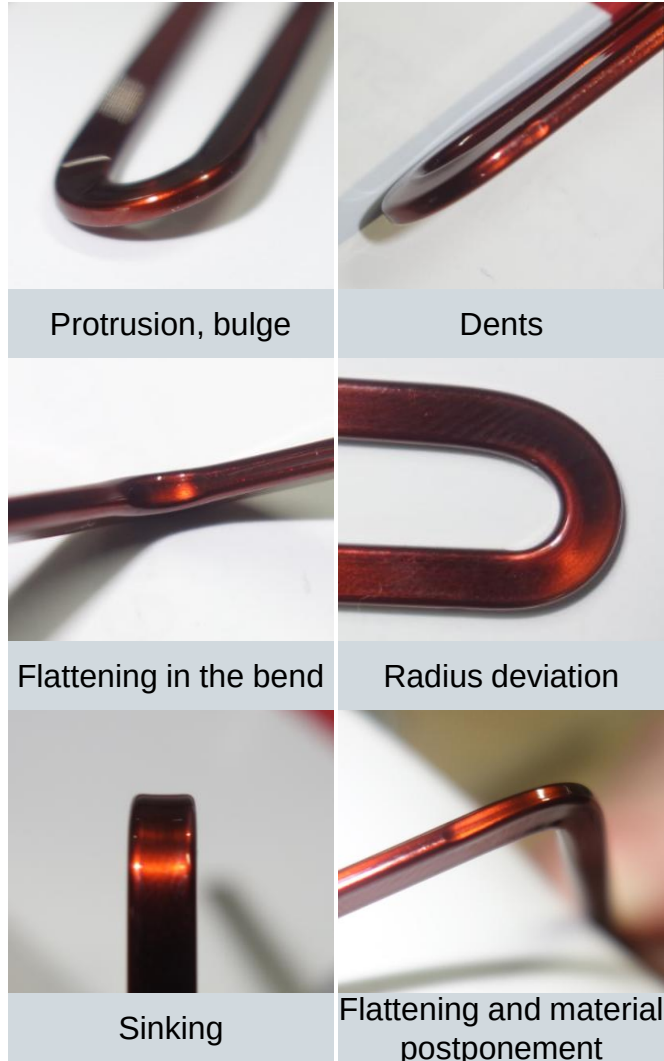


The production of open formed coils using the die bending process can be represented by a two-stage process.



Various methods are available for qualifying the electrical properties and bending quality.

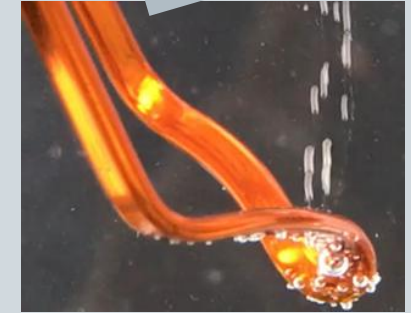
See lecture
unit F10



Pin-hole test

also: error count test

- Electrolysis bath for the visualization of damage to the insulation layer due to rising gas bubbles
- Copper wire as anode and aluminum sheet as cathode
- Standard: DIN EN 60851-5



Durability

- Immersion of the bent copper wire samples in a grist bath
- Up to 5.5 kV AC voltage (60 Hz)
- Standard: DIN EN 60851-5

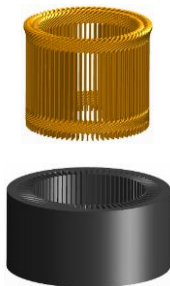


Partial discharge measurement

- Defined placement of a bent wire sample on a steel electrode
- Tracing of a voltage curve and recording of PDIV (inception voltage of the PD) and PDEV (drop-out voltage of the PD)
- Up to 5.5 kV AC voltage (60 Hz), no standardization



Open formed coils can be assembled by inserting them axially into the metal sheet stack.

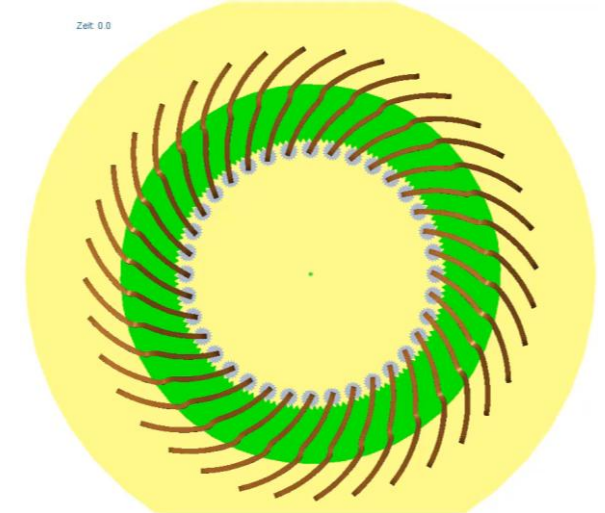
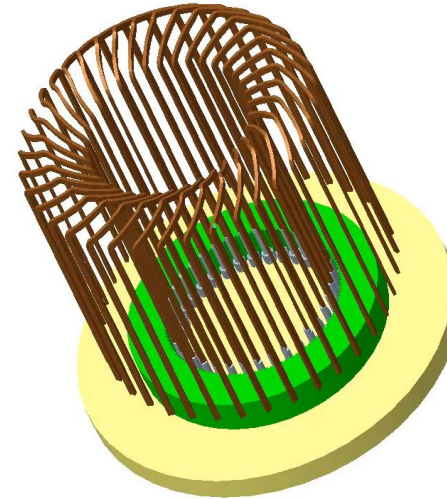
Requirements		■ Ensuring the correct arrangement of the formed coils ■ Low production and cycle times ■ No damage to the insulation layer ■ Reproducible assembly	
Individual assembly of the open formed coils	Advantages and disadvantages	■ Flexibility (++) ■ Cycle time (--) ■ Susceptibility to errors (--) ■ Process reliability (-) ■ Assembly effort (--)	 
Joining pre-assembly modules	Advantages and disadvantages	■ Cycle time (++) ■ Susceptibility to errors (+) ■ Process reliability (++) ■ Assembly effort (+) ■ Flexibility (--)	    

To produce the pre-assembly modules, the open formed coils are first inserted into a tool that performs a twisting process.

Production of the pre-assembly modules

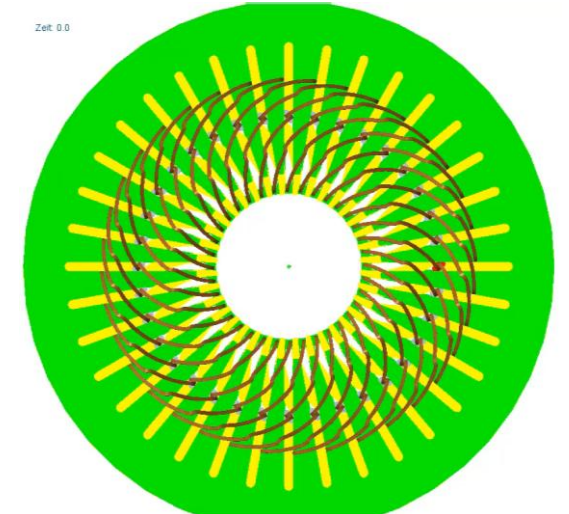
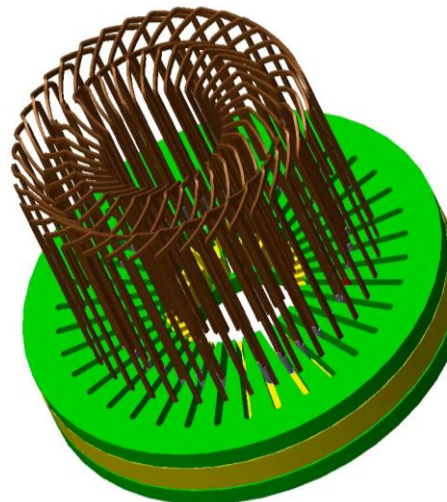
Assembly basket for one level

- Individual pins are placed on the tool
- Relative movement of the semi-finished products according to "basket assembly"
- "Turning in" the semi-finished products to form the basket
- Handling of the entire basket and assembly in the stator

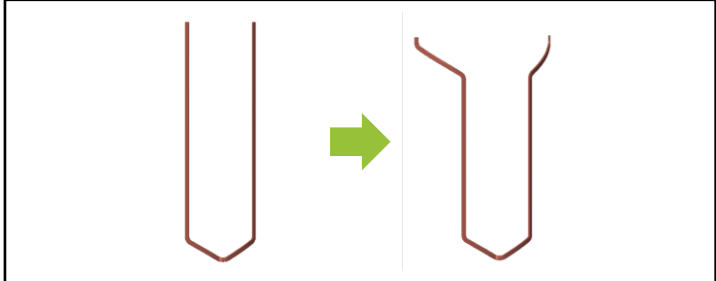
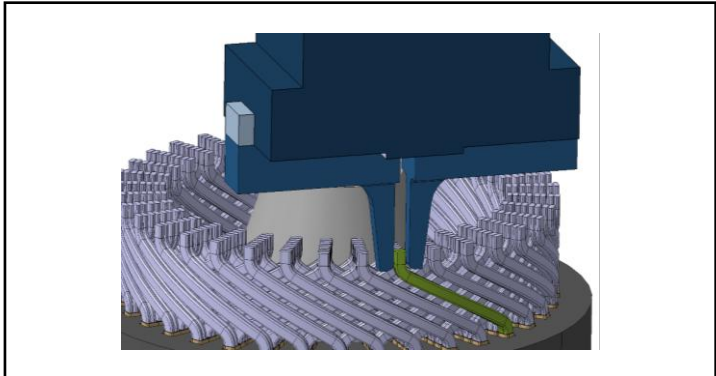
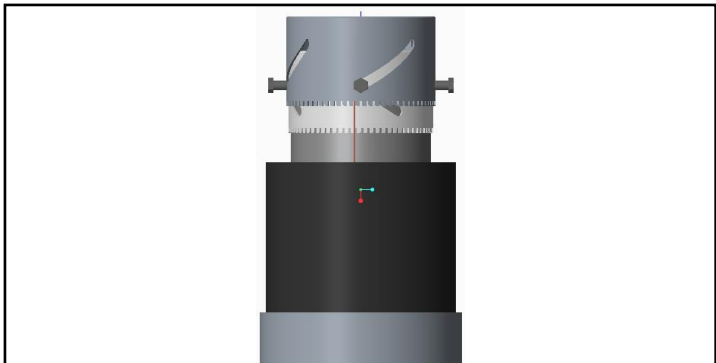


Multi-level assembly basket

- Individual pins are placed on the tool
- Synchronized "screwing in" of two layer levels
- Handling of the complete basket consisting of two layer levels and assembly in the stator



In the offset operation, called "twisting," the free conductor ends are set to allow contacting.

Requirements		- Precise positioning of the wires is essential - Wire springback must be considered - Undamaged condition of the paint insulation must be ensured	
Bending single wires	Advantages and disadvantages	- Flexibility (++) - Cycle time (--) - Degree of automation (--) - Repeatability (-) - Process reliability (-)	
Simultaneous bending	Advantages and disadvantages	- Cycle time (++) - Degree of automation (++) - Repeatability (+) - Process reliability (+) - Flexibility (--)	

In the contacting process step, the free conductor ends are electrically and mechanically connected.

Contacting technologies

Technology	Automation capability	Flexibility	Tool wear	Process reliability	Accessibility	Cycle time
Resistance welding	+	0	--	0	0	+
Laser welding	++	+	++	++	++	++
Ultrasonic welding	+	0	+	+	-	++
Crimping	++	-	+	+	0	++
Hot crimping	+	+	--	0	-	0
Ultrasonic crimping	+	+	+	+	-	0
Ultrasonic compacting	++	0	+	+	--	++
Soldering (hard/soft)	--	++	++	-	++	0
Cutting clamps	+	+	++	0	+	++

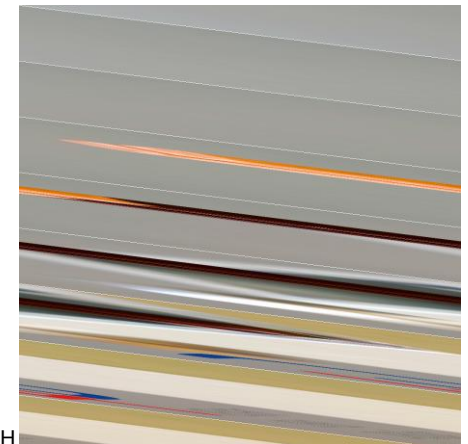


Contacting of formed coil or bar windings

- Number of contact points
- Cycle time
- Non-productive times
- Process reliability
- Weld spatter
- Mechanical stress
- Heat input
- Position tolerances
- Accessibility
- Connection cross-section
- Contact resistance
- Mechanical strength

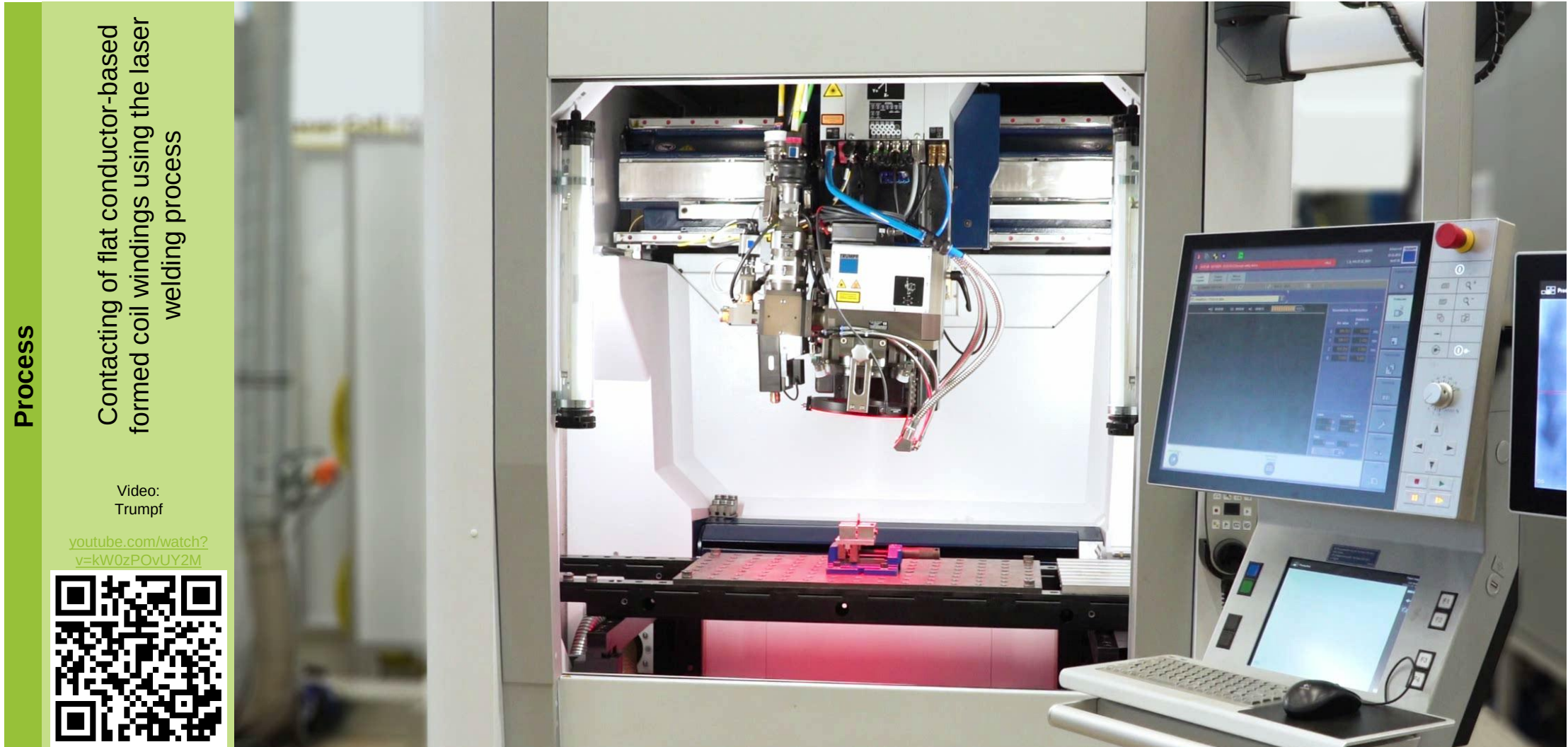
Potential of laser beam welding

- + Short cycle times
- + Beam guidance with scanner
- + Process stability
- + High welding depths possible
- + Contactless process
- Formation of ejections



Video: 4D GmbH

Laser beam welding of formed coil windings shows potential for producing the large number of contact points on formed coil windings.

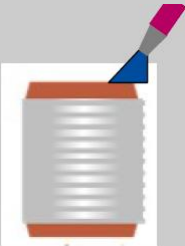
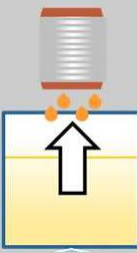
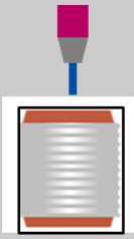
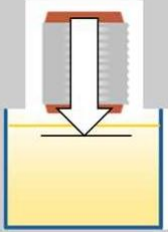


After the contacting process, the exposed contact points must be isolated.

See lecture
unit F8

Contacting		Tasks	Coating	Potting
		■ Insulating contact points to each other ■ Protection against moisture and dirt		

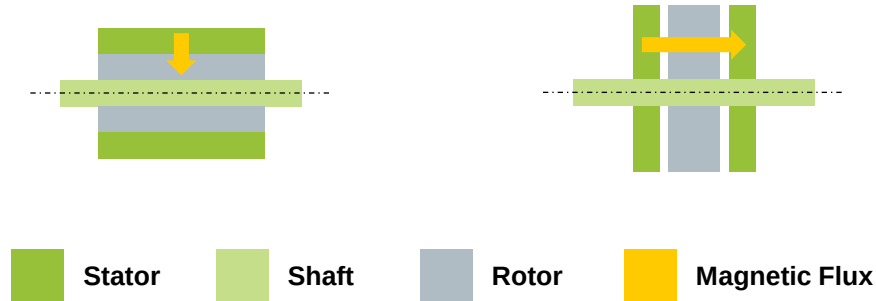
Technologies for insulating the contact points of formed coil/bar windings

	Spraying	Dripping	Current-UV	Potting	Dipping
Insulation materials	Resins	Resins	Resins	Resins	Powder or resins
Principle sketch					
Advantages	Simple procedure	Low stator contamination	High dielectric strength	High heat conduction	Simple procedure
Disadvantages	Thin insulation layer on edges	Good accessibility required	Contamination of complete stator	High mass increase	Low reproducibility

Source/Graphics: ELANTAS

The advantages in terms of weight, compact design, high torque and efficiency make the axial flux topology an interesting alternative to the radial flux motor.

Design of radial (left) and axial (right) flux motors



Radial flux motor

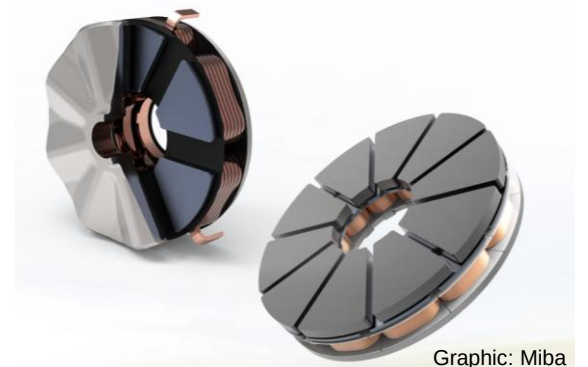
- Rotor and stator are formed by two concentric cylinders
- Magnetic flux between the two components is radial to the rotor axis

Axial flux motor

- Rotor and stator have a disc-like structure
- Magnetic flux between the two components is axial to the axis of rotation

Advantages of axial flux motors

- **Compact motor design:** Due to their design and operation, axial flux motors can be made shorter in the axial direction, making them more flexible for integration in various applications
- **High torque:** The motor structure enables a large effective air gap diameter, which leads to a correspondingly high torque
- **Reduced weight:** The specific design of the axial flux motor leads to better utilization of the materials and thus to a lower weight and improved power density
- **Higher efficiency:** The comparatively lower operating speed range results in lower frequency-dependent inverter losses



The contents of the lecture unit can be further deepened with the following specialized literature:

Reference books

- Much W.; et al (1986): „Wicklungen und Montage rotierender elektrischer Maschinen“, VEB Verlag Technik, Berlin, ISBN 3-341-00020-8.
- Bălă C.; et al. (1976): „Handbuch der Wickeltechnik elektrischer Maschinen“, VEB Verlag Technik, Berlin.
- Kampker, A. (2014): Elektromobilproduktion. Berlin: Springer Vieweg. ISBN 978-3-642-42022-1



The contents of the lecture unit can be further deepened with the following specialized literature:

Essays

- Riedel, A., Masuch M., Weigelt M., Gläsel T., Kühl A., Reinstein S. und Franke J. (2018): „Challenges of the Hairpin Technology for Production Techniques“. *2018 21st International Conference on Electrical Machines and Systems (ICEMS)*.
- Weigelt, M., Fröschl, C., Masuch, M., Riedel, A., Franke, J. (2018): „Erkenntnisse aus der Biegesimulation von Flachdraht“. In: *Draht* 69, 2018 (1/2018).
- Kühl A., Riedel A., Gläsel T., Masuch M. and Franke J. (2019): „Robot-based Production of Electric Motors with Hairpin Winding Technology“, *International Conference on Intelligent Automation and Robotics (ICIAR'19)*.
- Gläsel T., Bachinski Pinhal D., Masuch M., Kühl A., Franke J. and Gerling D. (2019): “Manufacturing Influences on the Motor Performance of traction Drives with Hairpin Winding”, in *2019 9th IEEE Electric Drives Production Conference (EDPC)*.



Dissertations

- Gläsel T. (2020): Prozessketten zum Laserstrahlschweißen von flachleiterbasierten Formspulenwicklungen für automobile Traktionsantriebe. Erlangen: FAU-Press.



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THANK YOU